# RAJALAKSHMI ENGINEERING COLLEGE

**(An Autonomous Institution)**

**Rajalakshmi Nagar, Thandalam – 602 105**

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

## CR23532 – CYBER AND DIGITAL FORENSICS

## LABORATORY RECORD

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| **Year / Semester :** | III Year / V Semester |
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# EXPERIMENT – 1

## Study of Computer Forensics and Different Tools Used for Forensic Investigation

### AIM

To gain knowledge of digital forensics, understand its role in investigating cybercrimes, and explore the various software tools used for collecting, examining, and analysing digital evidence.

### INTRODUCTION

Digital forensics is a branch of cybersecurity that deals with the identification, collection, preservation, examination, and analysis of digital evidence related to cyber incidents and computer-based crimes. It helps investigators understand how an incident occurred, identify possible sources of unauthorised activity, and gather evidence that may assist in an investigation.

During a cybercrime, traces of activity may remain on digital devices in the form of system logs, registry entries, malicious programs, file modifications, browser history, and remnants of deleted data. Digital forensic investigators examine these traces using specialised techniques to reconstruct events and obtain reliable evidence. Proper procedures are followed to maintain the integrity and authenticity of the collected information.

Several software applications are available for conducting digital forensic investigations. Commercial tools such as EnCase Forensic, Forensic Toolkit (FTK), and ProDiscover provide advanced features for analysing digital devices and evidence. Open-source alternatives, including The Sleuth Kit (TSK), Autopsy, Helix, and Knoppix-based forensic environments, also support forensic analysis. Additionally, distributions such as Kali Linux provide specialised utilities for examining filesystems, recovering data, analysing logs, and investigating security incidents.

These tools differ in their capabilities, interface, and intended use. Selecting an appropriate tool depends on the type of evidence, the investigation requirements, and the environment in which the examination is performed.

## PROCEDURE

1. Identified and studied commercial digital forensic software, including EnCase Forensic, Forensic Toolkit (FTK), and ProDiscover.
2. Examined open-source forensic solutions such as The Sleuth Kit (TSK), Autopsy, Helix, and Knoppix-based forensic environments.
3. Explored the forensic utilities available in Kali Linux and noted their applications in digital evidence examination.
4. Studied the features and purposes of commonly used forensic tools, including FTK Imager, Log Parser Lizard GUI, Noxcivis Field Toolkit, Active@ Partition Recovery, and Autopsy.
5. Compared the tools based on their major functions, such as evidence acquisition, file recovery, filesystem analysis, log examination, and digital investigation.
6. Organised the collected information into a table to summarise the purpose and applications of each forensic tool.

The function of each tool identified during the study is summarised in the table below.

**Forensic Tools Studied**

| Tool | Purpose |
|---|---|
| EnCase Forensic | Commercial forensic suite widely used by law enforcement. |
| FTK (Forensic Toolkit) | Commercial suite; FTK Imager creates forensically sound disk images. |
| ProDiscover | Commercial forensic investigation suite. |
| The Sleuth Kit (TSK) | Open-source collection of command-line forensic tools. |
| Autopsy | Graphical front-end for The Sleuth Kit; used to investigate disks and images. |
| FTK Imager | Forensic disk imager, memory capture and file recovery. |
| Log Parser Lizard GUI | Flexible and powerful log file parser. |
| Noxcivis Field Toolkit | Collects information from a live computer during incident response. |

| | |
|---|---|
| Active@ Partition Recovery | Recovers deleted and lost partitions. |
| USBDeview | Lists connected and previously used USB devices. |
| LastActivityView | Displays a log of recent user actions and system events. |
| ChromeHistoryView / IECacheView | Reads browser history and cache artifacts. |
| Compare It | Compares two files and highlights the differences. |
| DumpIt | Generates a physical memory dump of Windows machines. |
| Bulk Extractor | Extracts features such as e-mail, URLs and phone numbers from an image. |
| ExifReader | Extracts EXIF metadata from image files. |
| Recover My Files | Recovers deleted files from drives and disk images. |

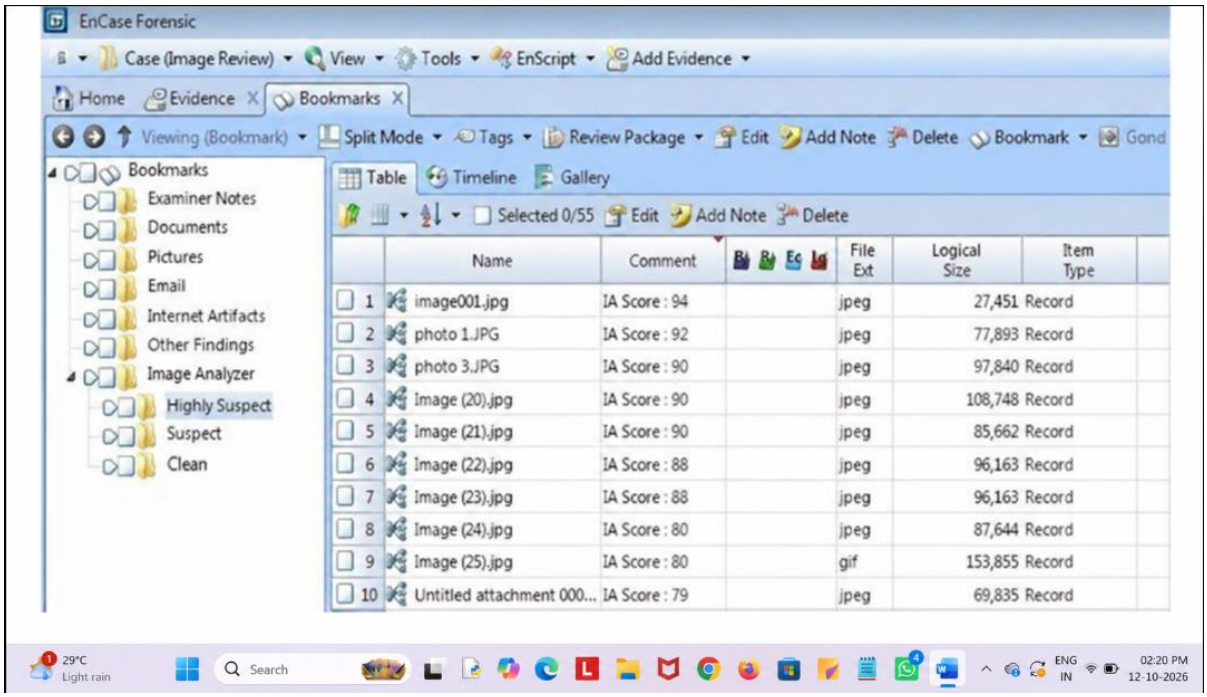



**Fig. 1.1** EnCase Forensic – a widely used commercial forensic suite

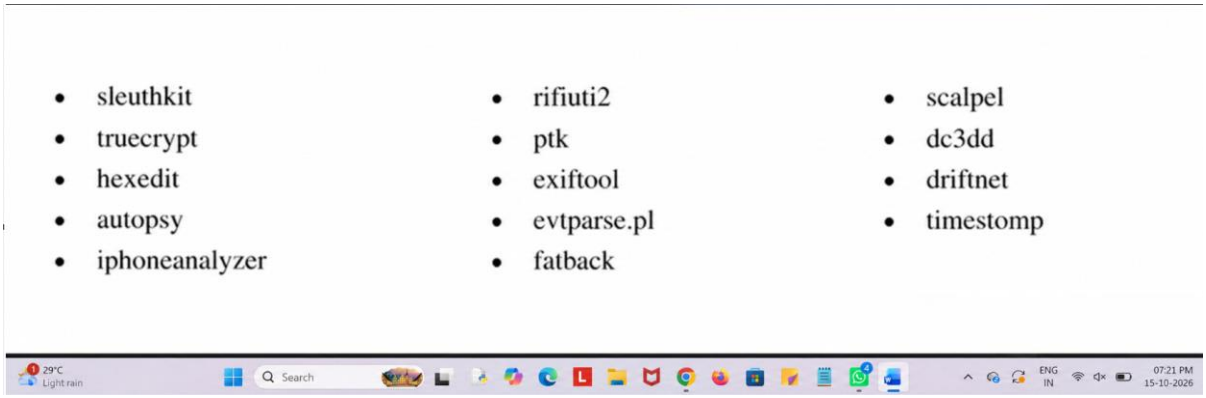



**Fig. 1.2** Some of the forensic tools available in the BackTrack / Kali distribution

**RESULT**

Thus, the field of computer forensics and the various tools used for forensic investigation were studied successfully, along with their purpose and usage.

# EXPERIMENT – 2

## Introduction to Digital Forensics (TryHackMe)

### AIM

To gain practical knowledge of digital forensics and learn the various stages of a forensic investigation by completing the "Intro to Digital Forensics" room on TryHackMe.

### PROCEDURE

- Logged into the TryHackMe platform (tryhackme.com) and accessed the "Intro to Digital Forensics" learning room.
- Explored Task 1, *Introduction to Digital Forensics*, to understand the concept of digital forensics and its applications in criminal investigations, corporate security incidents, and civil disputes.
- Studied Task 2, *Digital Forensics Process*, and learned the major stages involved in an investigation, including identification, preservation, collection, examination, analysis, and presentation of findings.
- Completed Task 3, *Practical Example of Digital Forensics*, by examining a sample scenario and understanding how digital evidence can be traced and interpreted.
- Attempted the questions provided in each task and used the available hints to clarify concepts and verify the answers.
- Reviewed the concepts covered in the room and checked the completion status after finishing all the required tasks.
- Summarised the key learnings related to digital evidence handling, forensic investigation procedures, and the importance of maintaining evidence integrity**.**

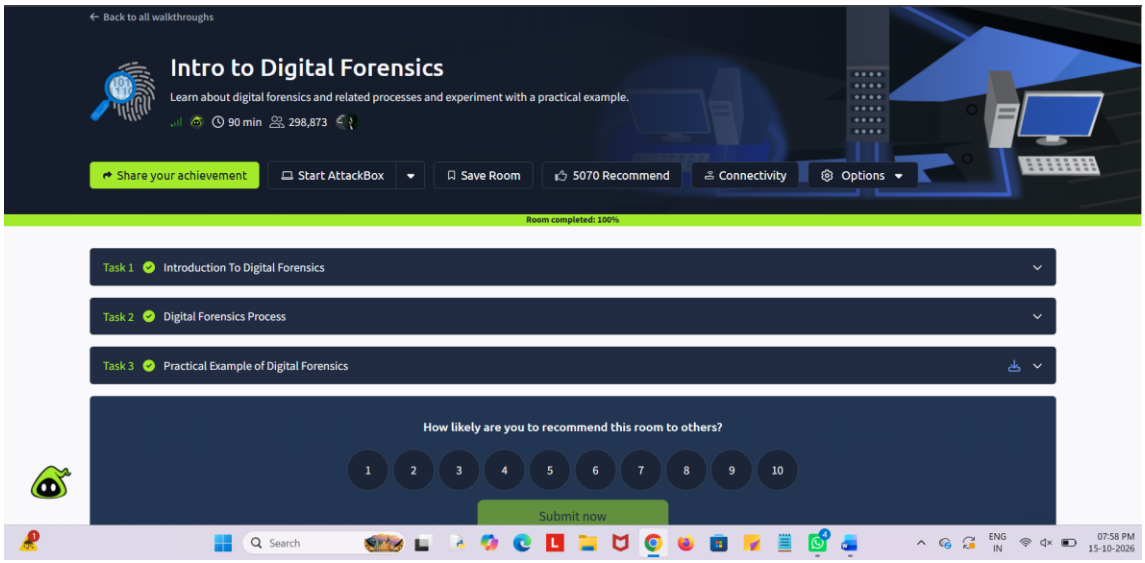



**Fig. 2.1** TryHackMe – "Intro to Digital Forensics" room showing 100% completion

## RESULT

The "Intro to Digital Forensics" room was completed with 100% progress. The basic concepts of digital forensics and the standard digital forensics process were understood.

# EXPERIMENT – 3

## Windows Forensics 1 (TryHackMe)

**AIM**

To study Windows registry forensics and understand the significant forensic artifacts available in the Windows operating system by completing the "Windows Forensics 1" room on TryHackMe.

**PROCEDURE**

- Logged into the TryHackMe platform and accessed the "Windows Forensics 1" room.
- Explored the *Introduction to Windows Forensics* section to understand the importance of Windows systems as sources of digital evidence and the different artifacts they retain.
- Studied *Windows Registry and Forensics* to learn about the major registry hives, including SAM, SYSTEM, SOFTWARE, SECURITY, NTUSER.DAT, and USRCLASS.DAT, along with the information stored in each hive.
- Examined the *Accessing Registry Hives Offline* section to understand how registry files acquired from a system image can be accessed and analysed using tools such as Registry Explorer.
- Learned about *Data Acquisition* and the methods used to collect relevant registry hives from live and offline Windows systems.
- Explored *Windows Registry* and *System Information and System Accounts* to identify artifacts related to the computer name, shutdown activity, user accounts, and logon details.
- Studied *Usage or Knowledge of Files/Folders* and *Evidence of Execution* to understand artifacts such as recently accessed files, ShellBags, Run keys, Prefetch files, and Jump Lists.
- Investigated the *External Devices / USB Device Forensics* section to learn how Windows registry artifacts can reveal information about previously connected USB devices. Completed the practical challenge provided in the room and reviewed the conclusion

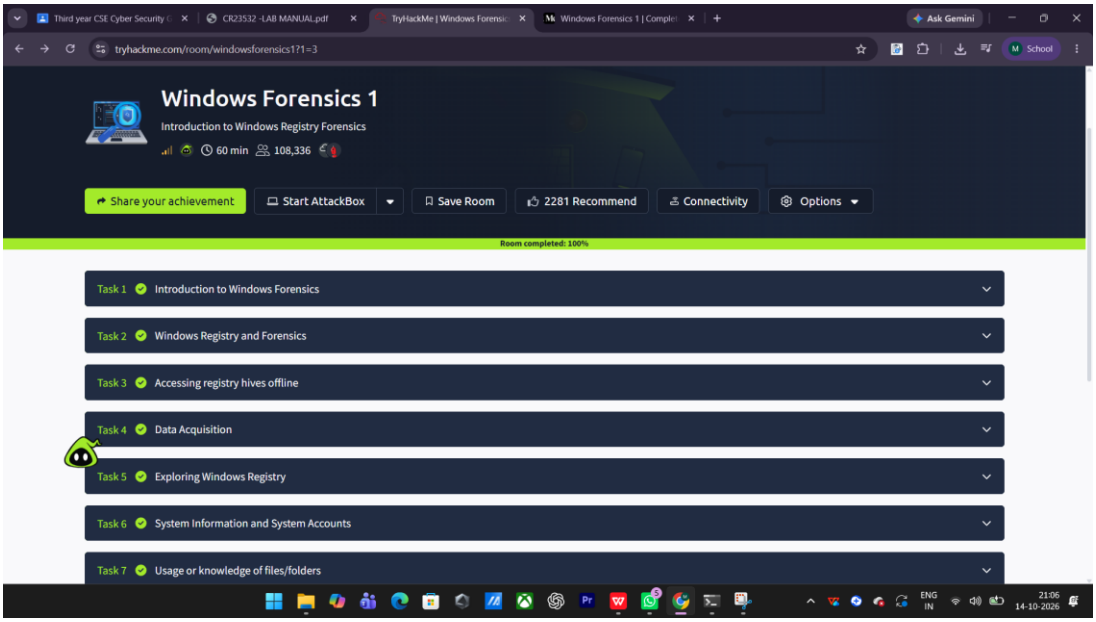



**Fig. 3.1** TryHackMe – "Windows Forensics 1" room with the list of tasks

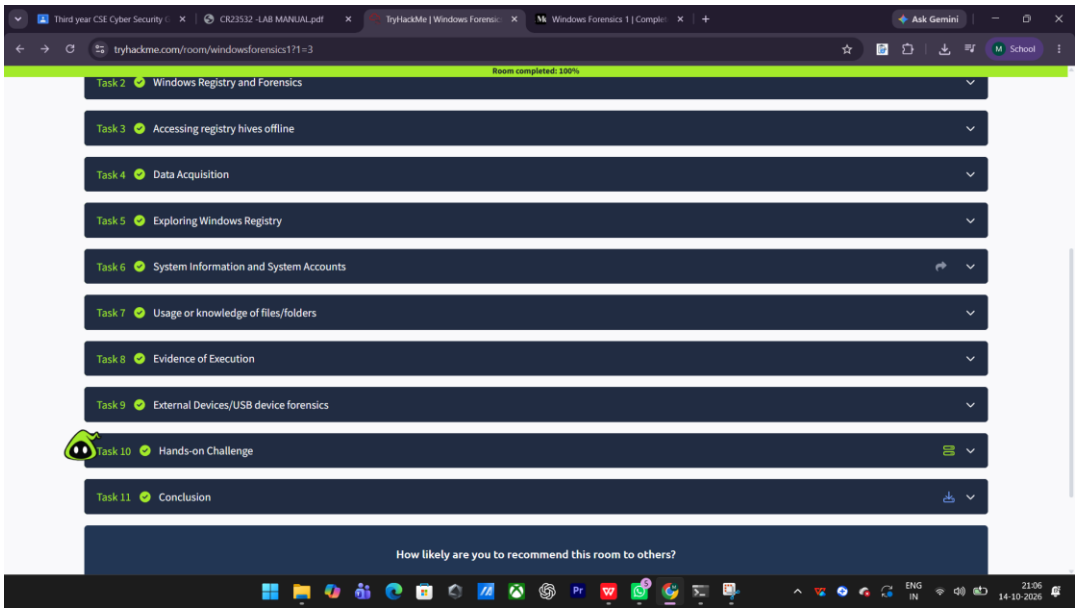



**Fig. 3.2** TryHackMe – Windows Registry and Forensics and evidence of execution sections

## RESULT

The "Windows Forensics 1" room was completed successfully. Windows registry hives, evidence of execution, file and folder usage artifacts, and USB device forensics were studied.

# EXPERIMENT – 4

## Live Forensics Case Investigation using Autopsy

### AIM

To investigate a disk image using the Autopsy digital forensic platform, analyse the extracted digital evidence, and prepare a forensic report containing the investigation findings.

### PROCEDURE

- Downloaded and installed Autopsy from its official website and launched the application to access the case management interface.
- Selected the *New Case* option and entered the case name and base directory to specify where the case database and related files would be stored.
- Provided the case number and examiner details, including the name, phone number, and email address, and completed the case creation process.
- Opened the *Add Data Source* wizard, selected the appropriate host, and chose the required data source type, such as a disk image, local disk, logical files, or unallocated space image.
- Browsed and selected the disk image or storage drive for examination, configured the input time zone as Asia/Calcutta, and specified the relevant image settings.
- Configured the available ingest modules, including Hash Lookup, File Type Identification, Extension Mismatch Detector, Embedded File Extractor, Picture Analyzer, Email Parser, Encryption Detection, Interesting Files Identifier, PhotoRec Carver, and Virtual Machine Extractor, according to the investigation requirements.
- Started the data processing by clicking *Finish*. Autopsy analysed the selected source and stored the extracted information and results in the case database.
- After processing was completed, used the *Generate Report* option to create a forensic report in a supported format, such as HTML or Excel, based on the available reporting modules.
- Examined the generated results to identify relevant forensic artifacts, including connected devices, EXIF metadata, installed applications, operating system details, user accounts, recently accessed documents, web bookmarks, cookies, downloads, browsing history, search activity, and email addresses.
- Reviewed the extracted evidence and report to understand the findings and document the information relevant to the forensic investigation.

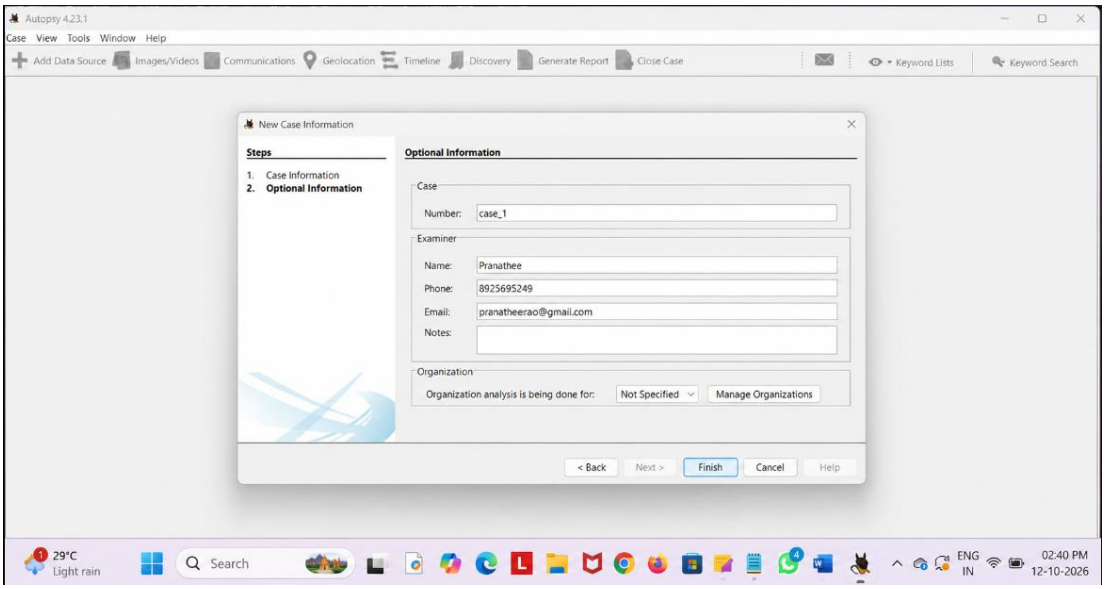



**Fig. 4.1** Autopsy – entering the examiner and optional case information

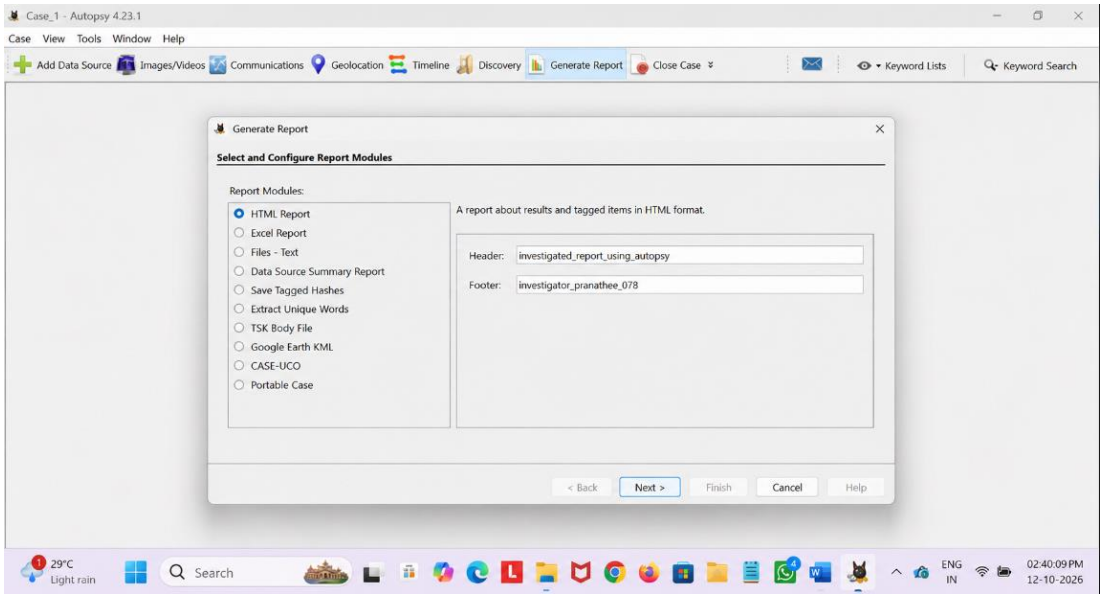



**Fig. 4.2** Autopsy – selecting and configuring the report modules

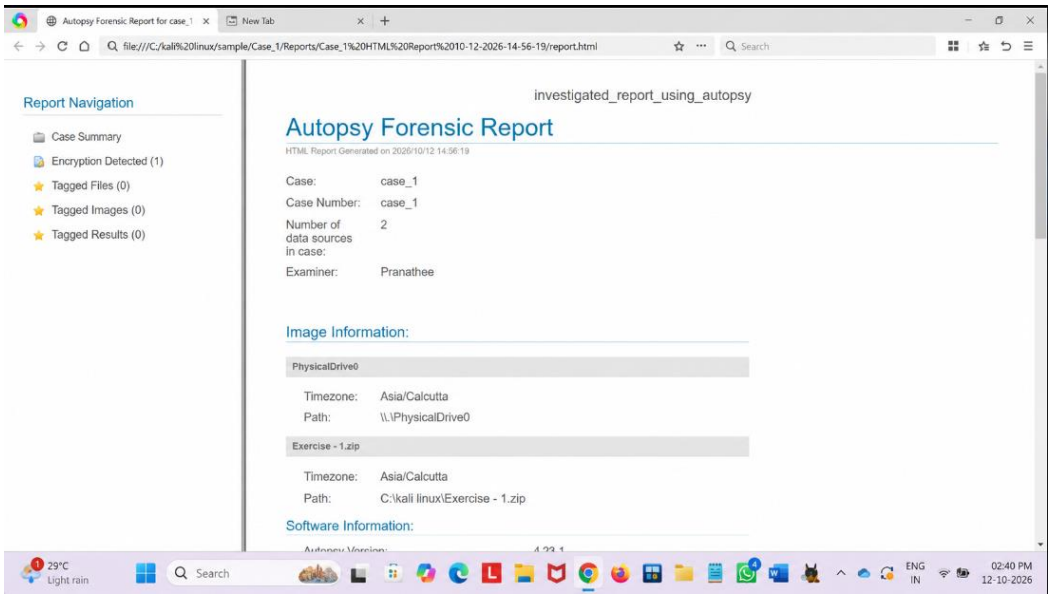



**Fig. 4.3** Autopsy forensic report – case summary and data sources

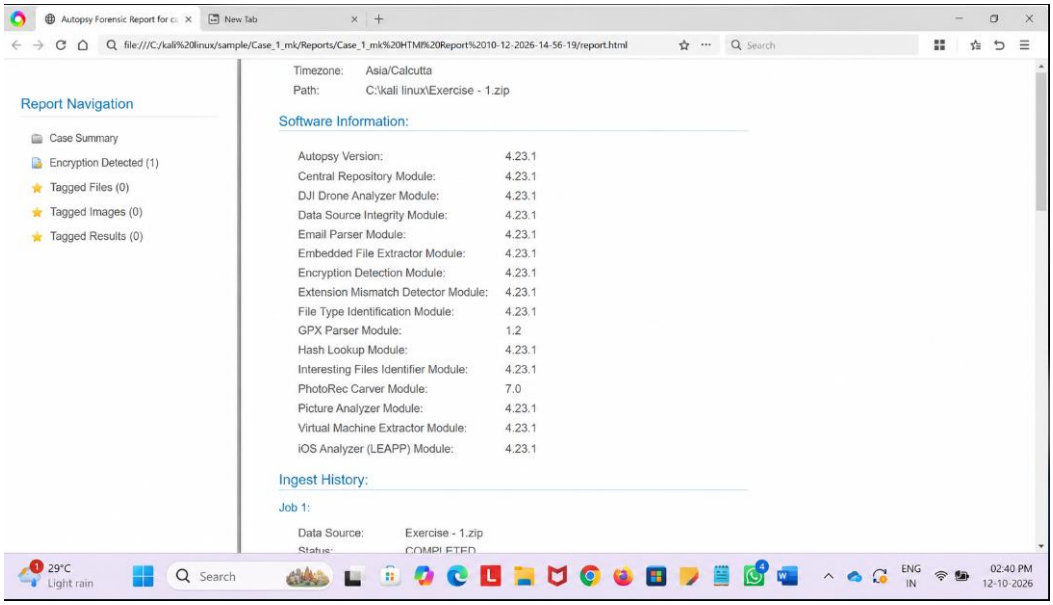



**Fig. 4.4** Autopsy report – software information and ingest modules

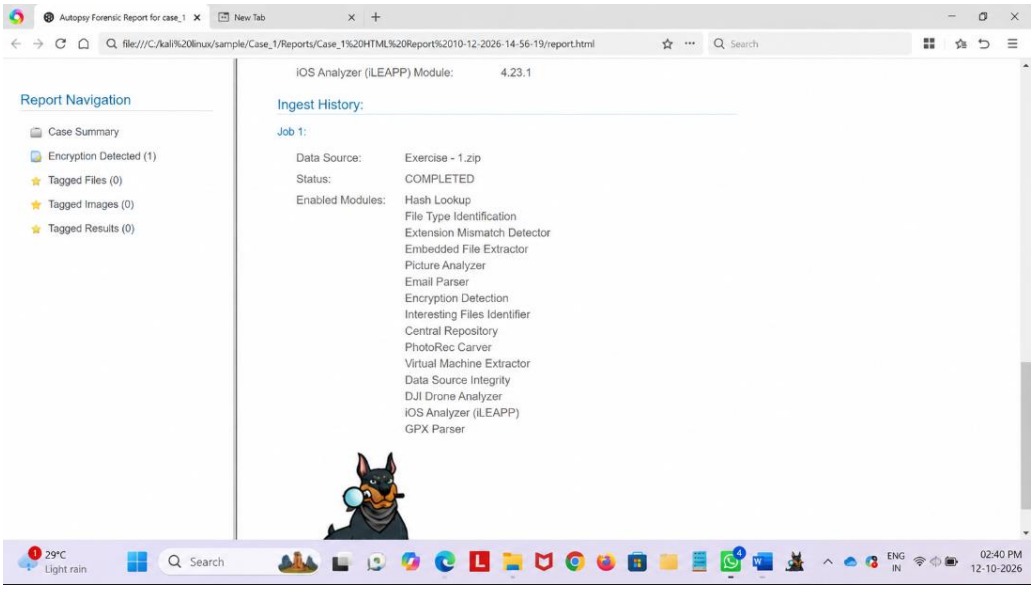



**Fig. 4.5** Autopsy report – ingest history and enabled modules

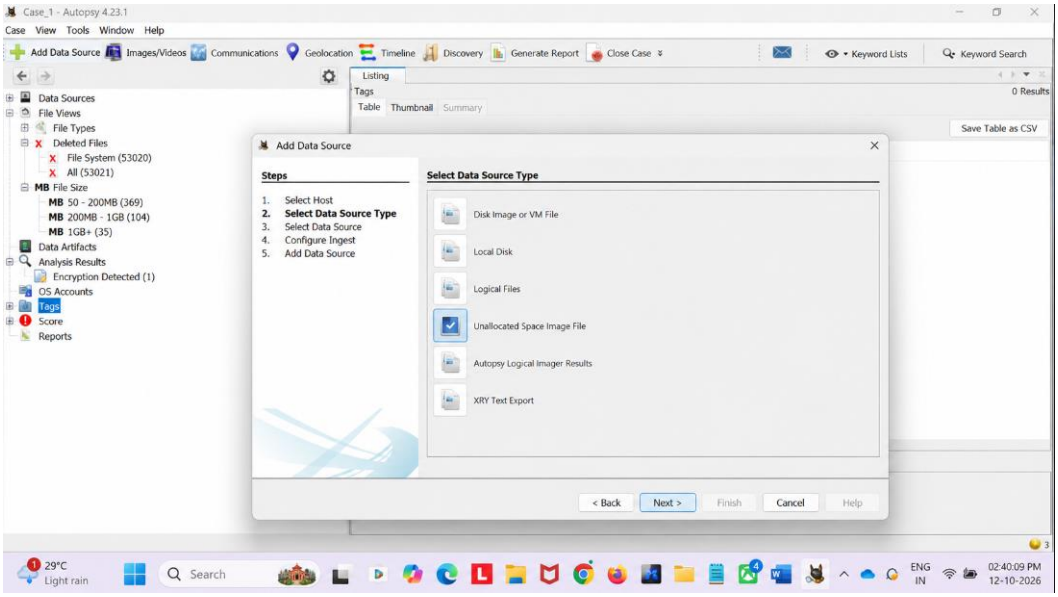



**Fig. 4.6** Autopsy – adding a data source (selecting the source type)

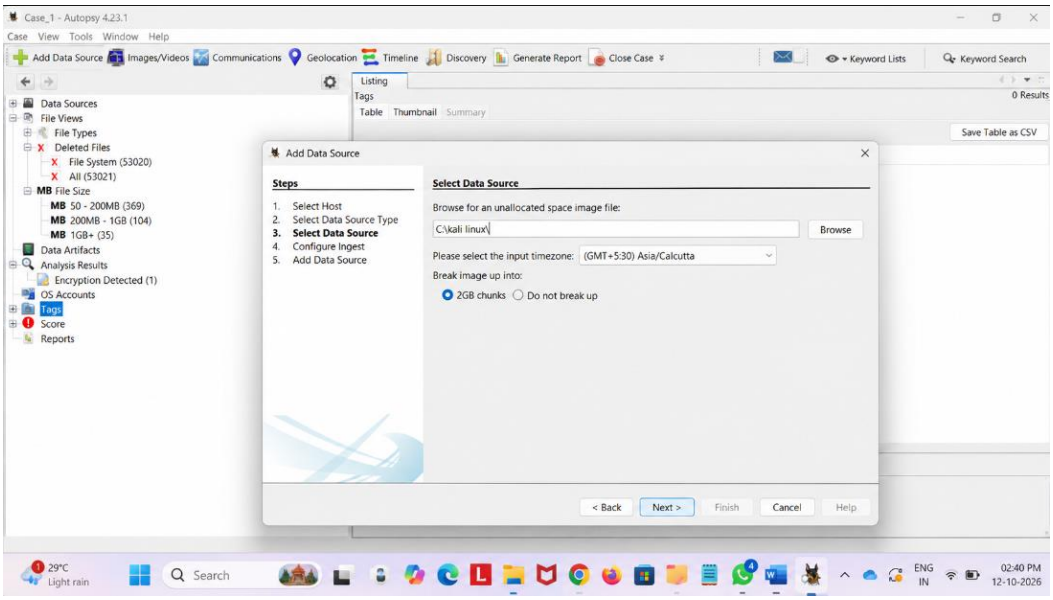



**Fig. 4.7** Autopsy – browsing for the disk image and setting the input time zone

## RESULT

Thus, a live forensic case investigation was carried out using Autopsy. The evidence device, the operating system information, user accounts, recent documents and browser artifacts were identified and the forensic report was generated successfully.

# EXPERIMENT – 5

## Recover Deleted Files using Forensics Tools (FTK Imager)

### AIM

To retrieve a deleted file from a storage device by acquiring a bit-by-bit forensic image using FTK Imager and validating the integrity of the acquired image through hash verification.

### PROCEDURE

- Step 1 – Create a sample file: Open Notepad, create a text file named malicious.txt, enter sample content, and save the file to the selected storage device.
- Step 2 – Delete the file: Delete the created file from the storage device. The file data may remain in the unallocated space until it is overwritten by new data.
- Step 3 – Understand forensic imaging: Recognise that normal file copying may not preserve deleted data. Therefore, a bit-by-bit forensic image is created to capture the relevant data for further examination.
- Step 4 – Launch FTK Imager: Install and open FTK Imager, navigate to the File menu, and select Create Disk Image.
- Step 5 – Select the source device: In the source selection window, choose the appropriate evidence source, such as a physical drive, logical drive, or an existing image file, and proceed to the next step.
- Step 6 – Select the image format: Choose a suitable output format, such as Raw (dd), SMART, E01, or AFF, depending on the available options and investigation requirements. The E01 format supports metadata and compression.
- Step 7 – Enter evidence information: Provide the necessary case details, including the case number, evidence number, evidence description, examiner's name, and additional notes.
- Step 8 – Configure the destination: Select the output folder, specify the image filename, and configure the fragment size and compression settings where applicable.
- Step 9 – Acquire the forensic image: Click Start to begin the imaging process. Monitor the progress until FTK Imager completes the acquisition.
- Step 10 – Verify image integrity: Use the Verify Drive/Image option, where applicable, to calculate and compare the relevant hash values, such as MD5 and SHA-1. Matching hashes provide evidence that the imaged data has remained unchanged during acquisition.
- Step 11 – Recover the deleted file: Examine the acquired image using suitable forensic tools to locate and recover the deleted file, if its contents remain available. Compare the recovered content with the original sample to assess the success of the recovery process.

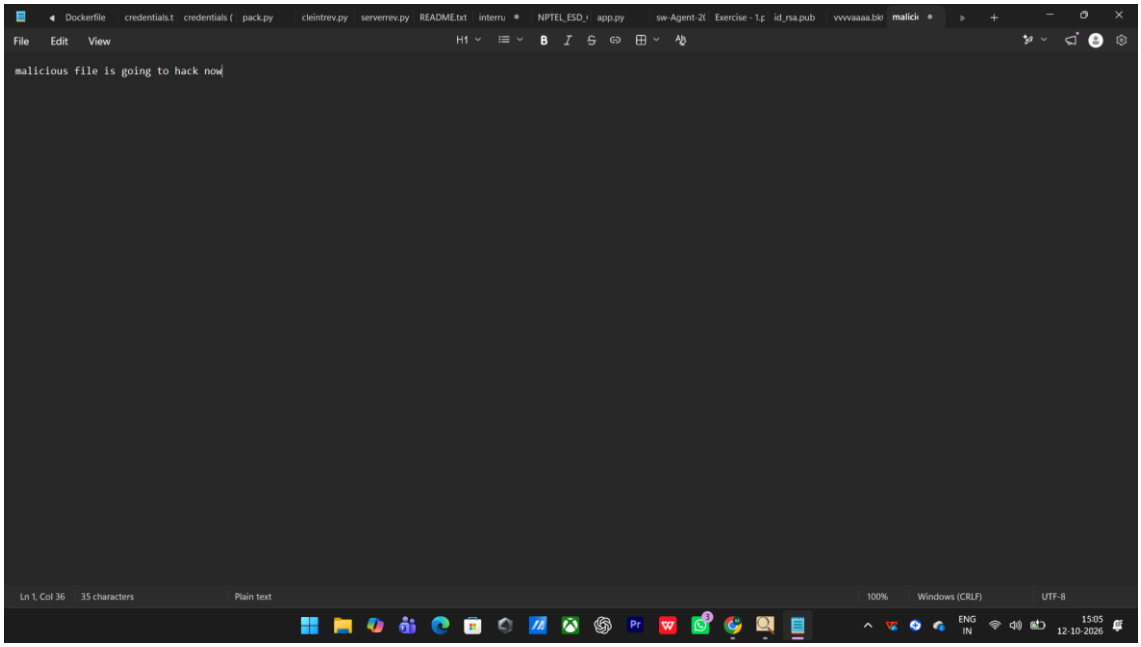



**Fig. 5.1** Creating the target file 'malicious.txt' with the sample content

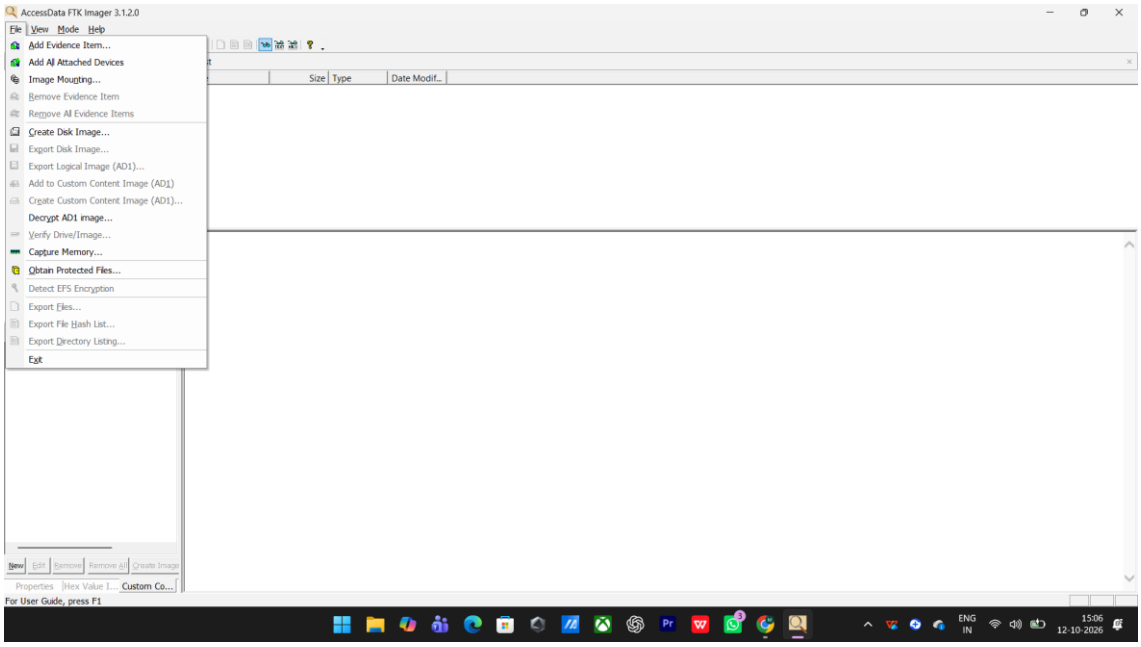



**Fig. 5.2** FTK Imager – selecting 'Create Disk Image' from the File menu

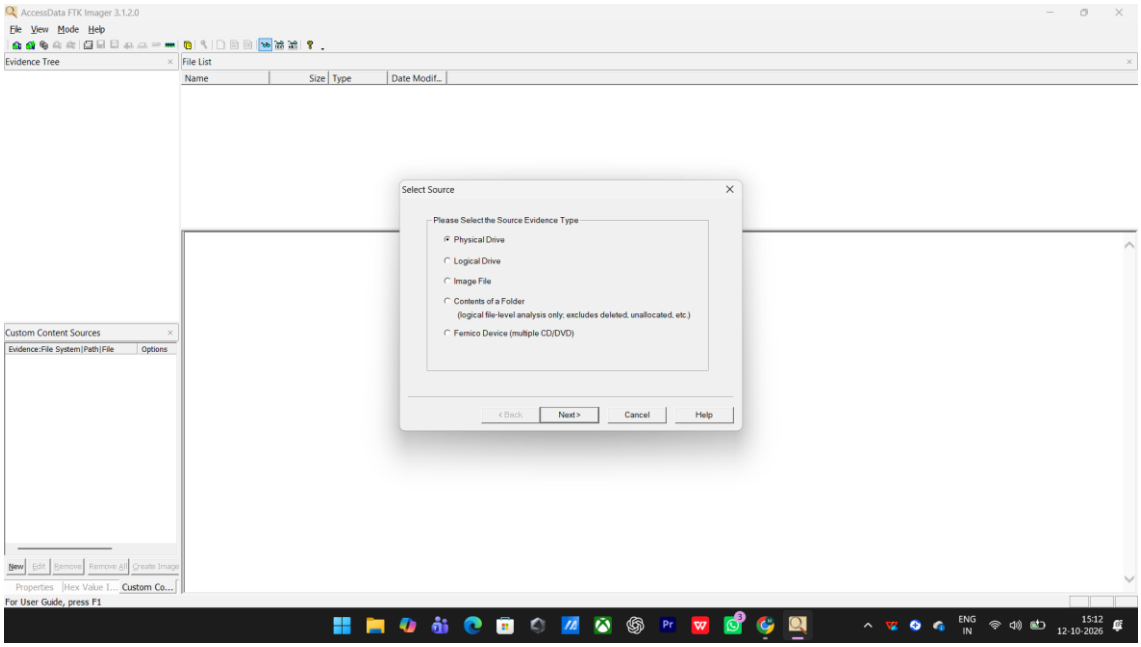



**Fig. 5.3** FTK Imager – selecting the source evidence type

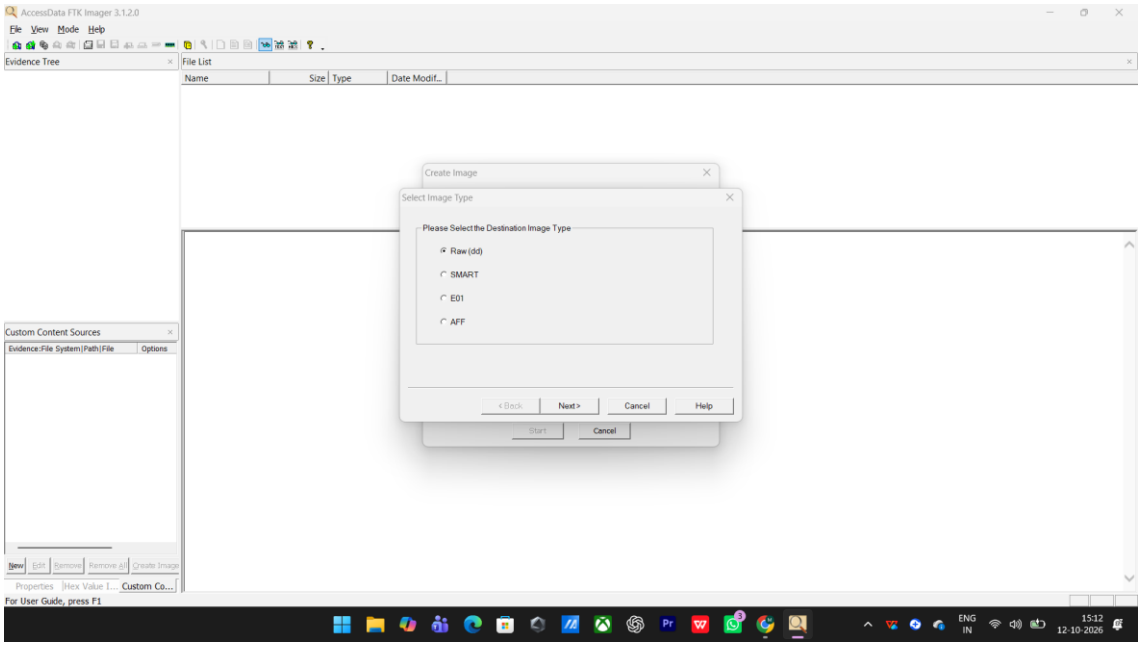



**Fig. 5.4** FTK Imager – choosing the destination image type (Raw dd)

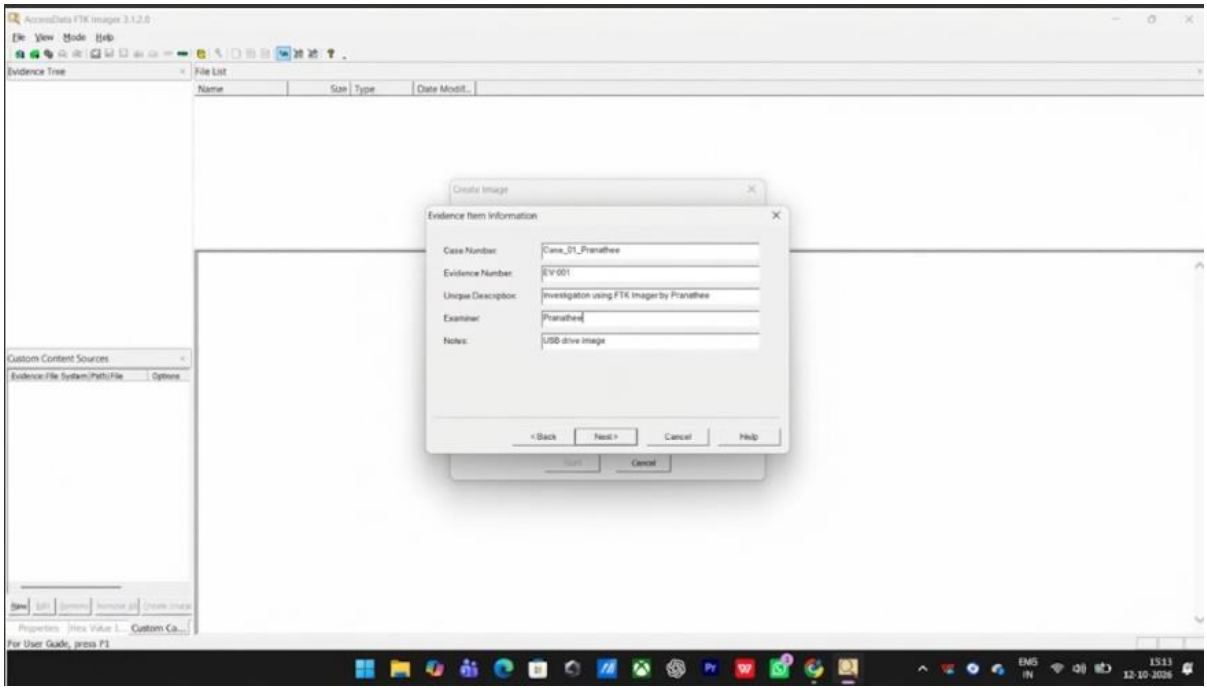



**Fig. 5.5** FTK Imager – entering the evidence item information

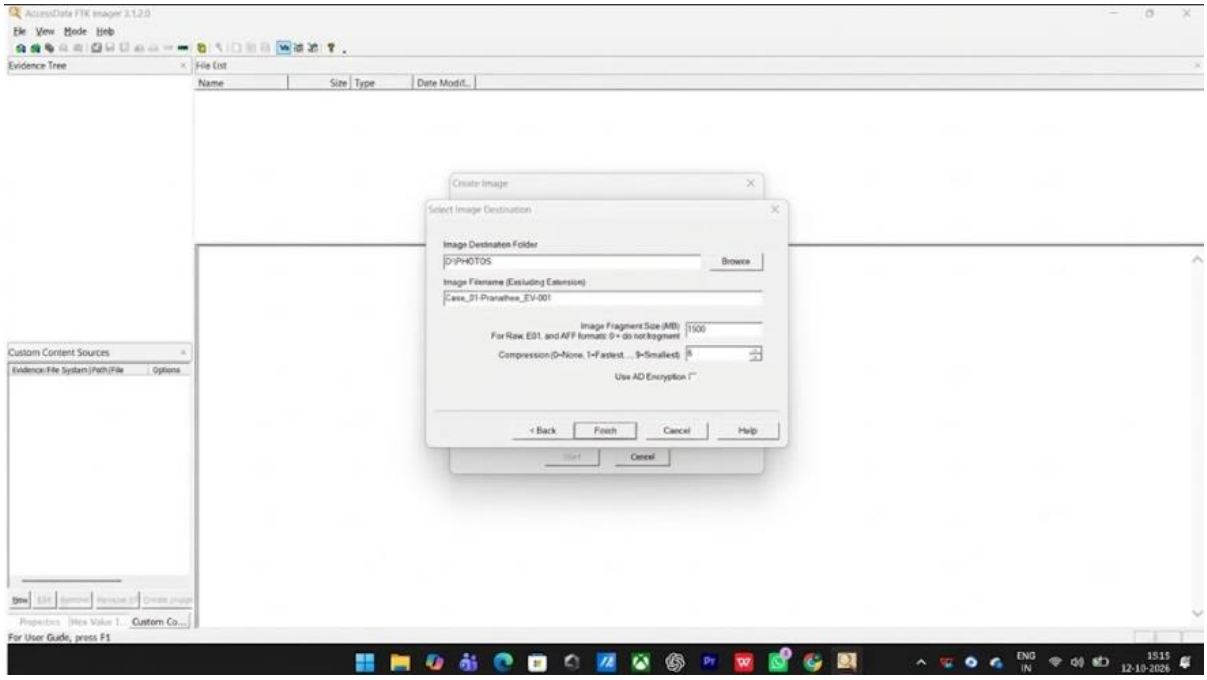



**Fig. 5.6** FTK Imager – specifying the image destination folder and file name

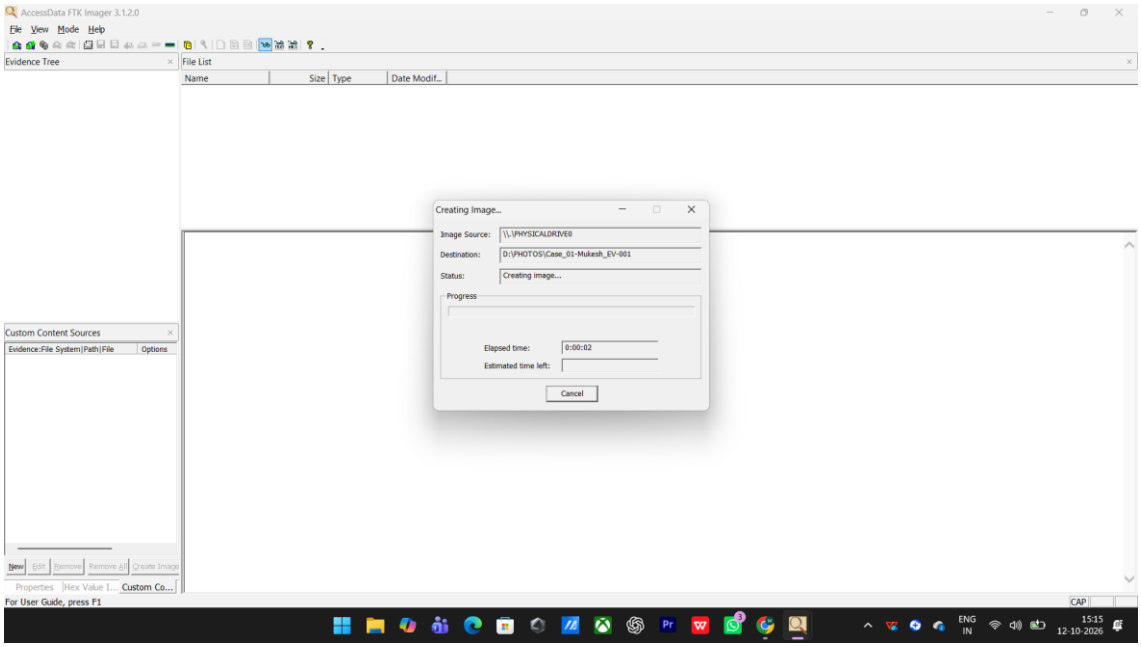



**Fig. 5.7** FTK Imager – creating the bit-by-bit image of the drive

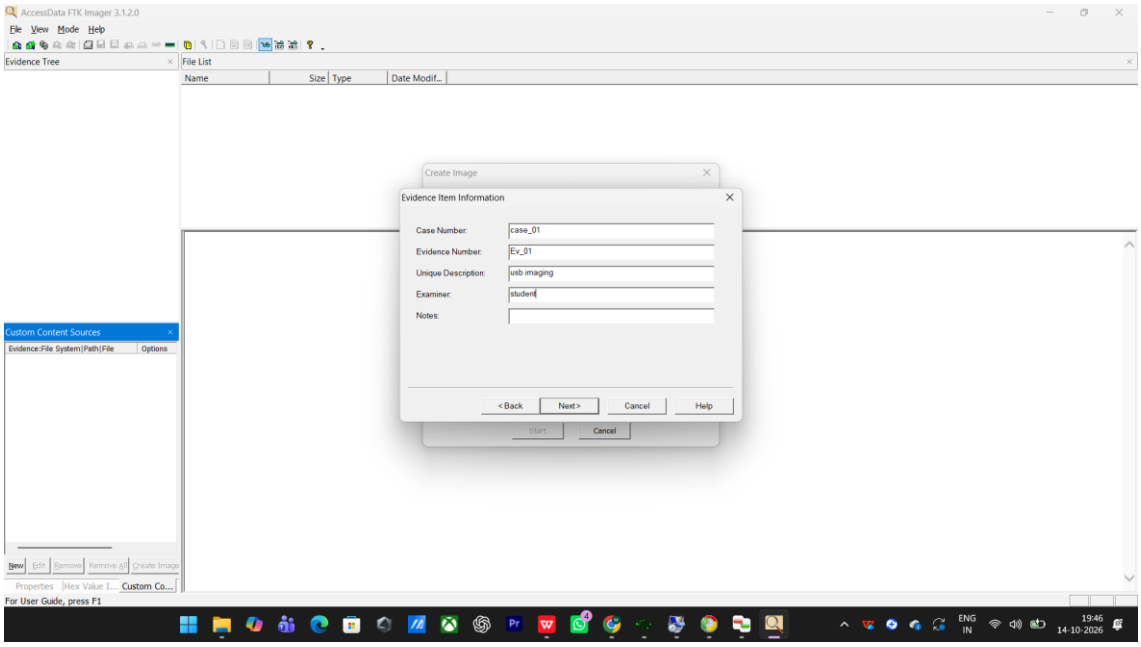



**Fig. 5.8** FTK Imager – entering the evidence item information for the device image

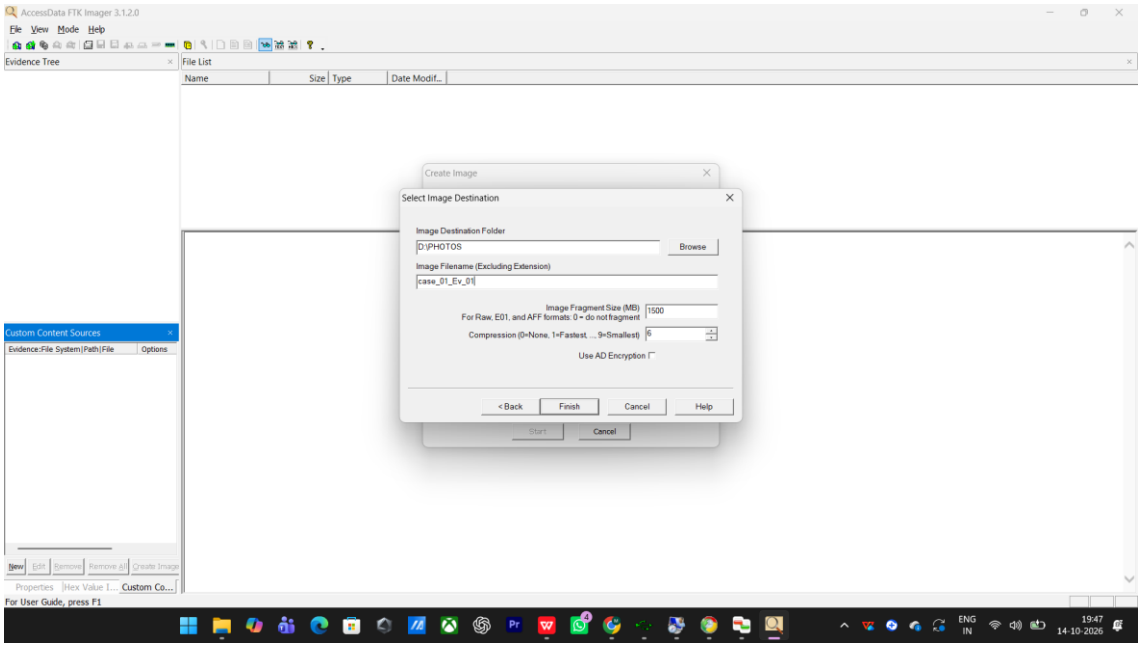



**Fig. 5.9** FTK Imager – setting the image destination folder

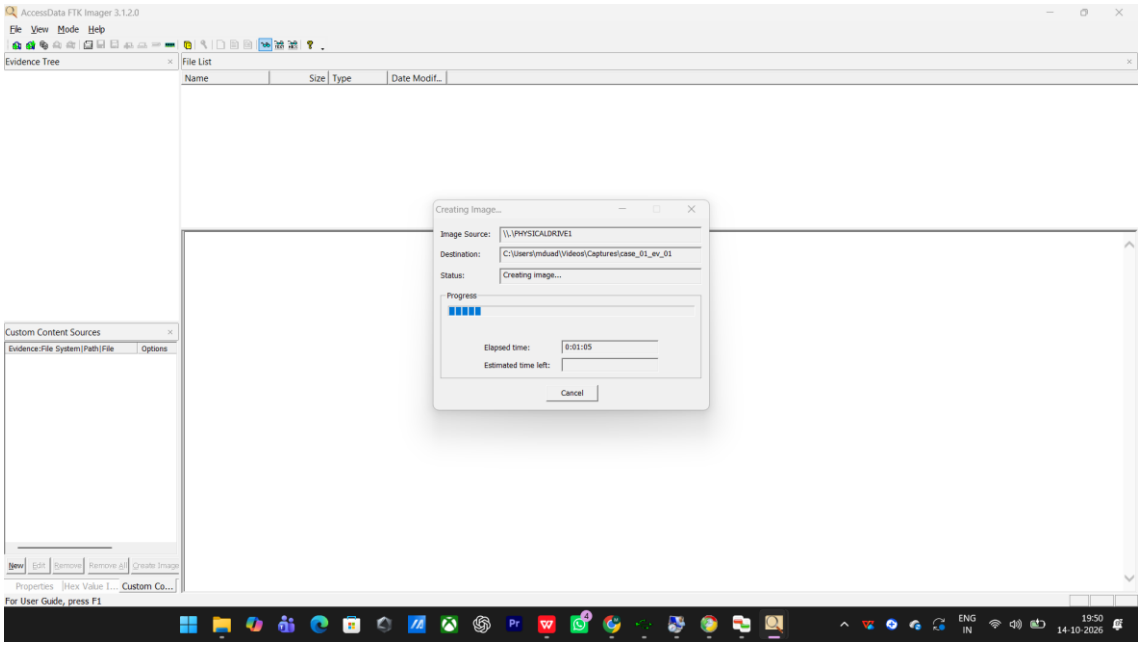



**Fig. 5.10** FTK Imager – imaging the device (creating the image)

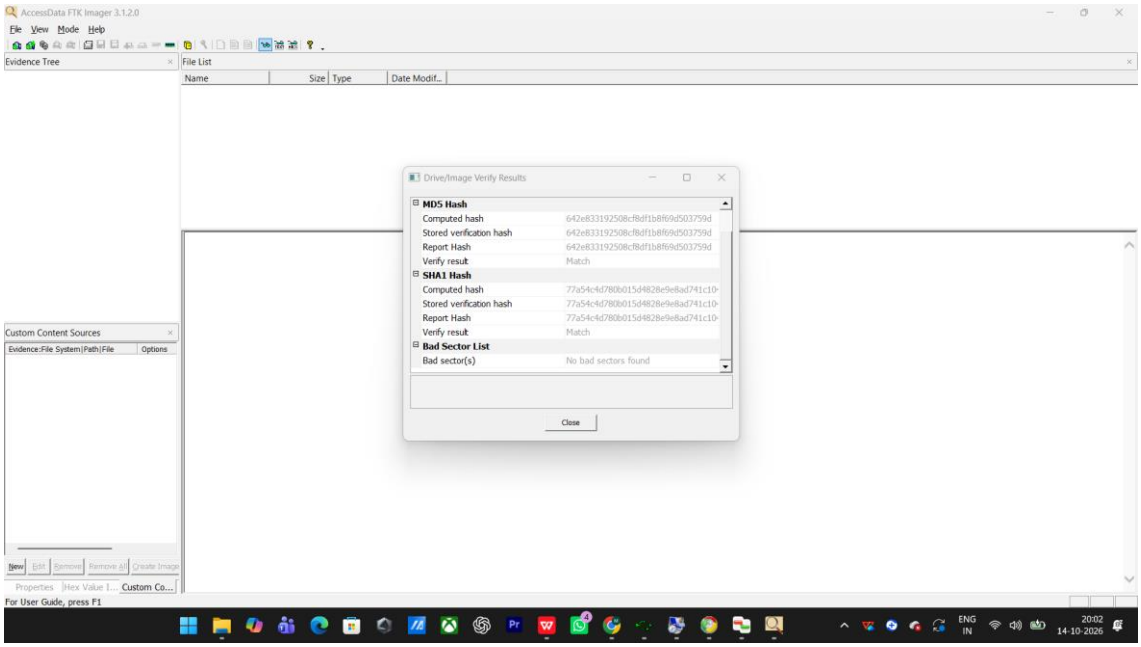



**Fig. 5.11** FTK Imager – verifying the image (MD5 and SHA-1 hash values)

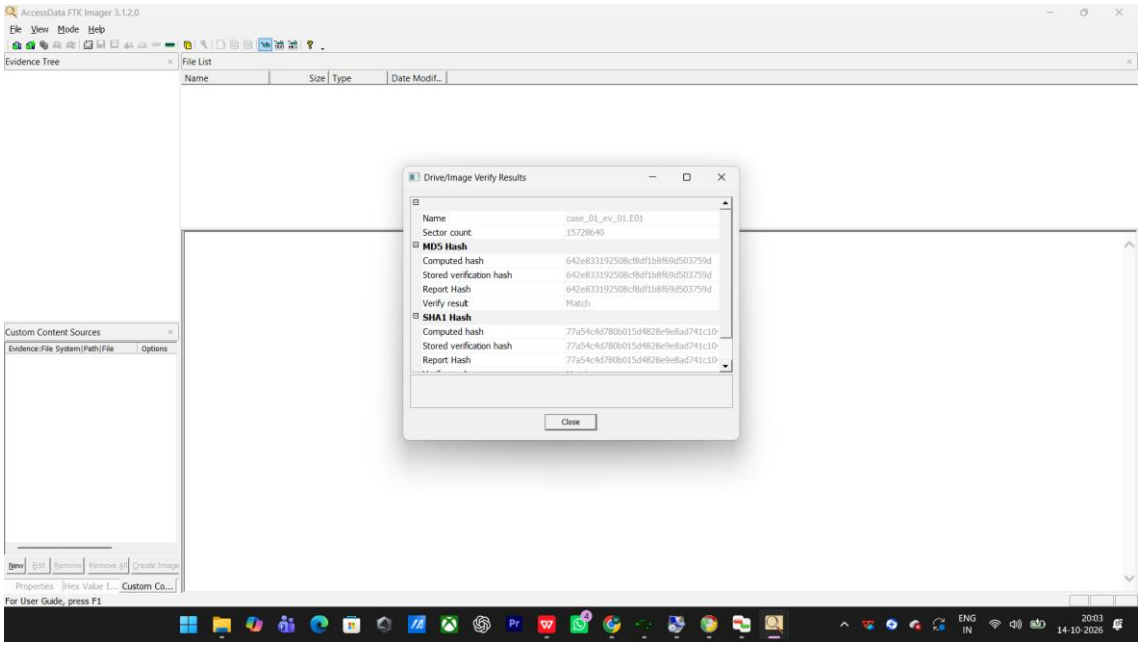



**Fig. 5.12** FTK Imager – drive/image verification results

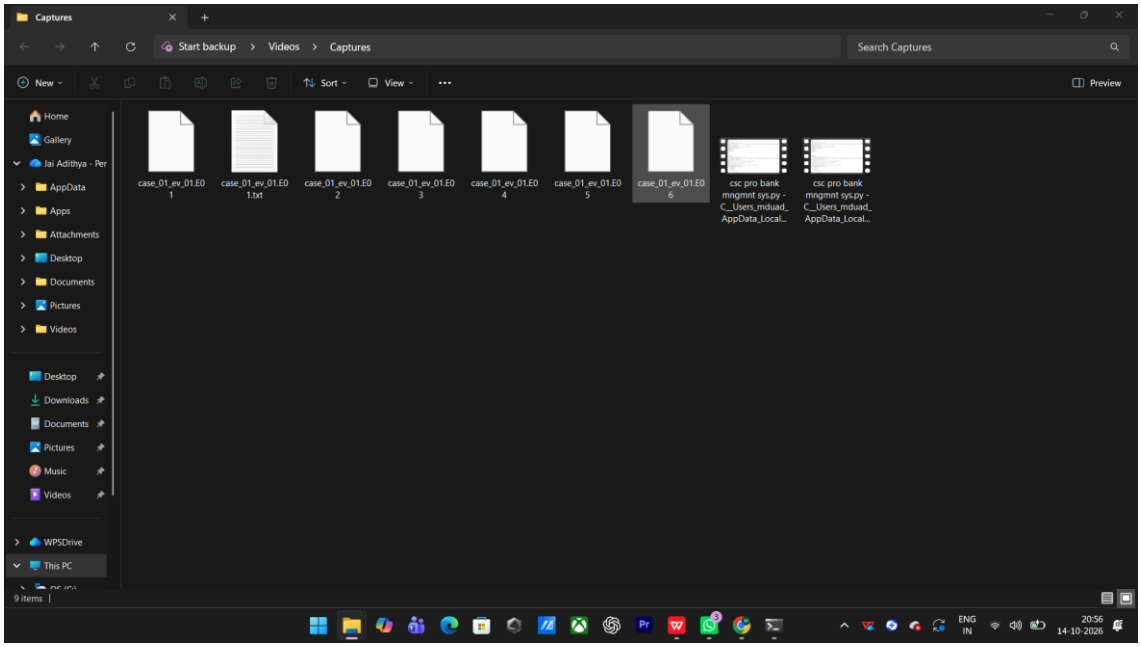



**Fig. 5.13** The generated E01 image segments of the device

## RESULT

Thus, the deleted file was recovered using forensic tools. A bit-by-bit image of the device was created with FTK Imager, the integrity of the image was verified using the MD5 and SHA-1 hash values, and the deleted content was recovered and verified successfully.

# EXPERIMENT – 6

## Find Last Connected USB on the System (USB Forensics)

### AIM

To identify the last connected USB device on the system using USBDeview.

### PROCEDURE

1. Download USBDeview from the official NirSoft website and run the application on the system.
2. Observe the list of currently connected and previously used USB devices displayed by the tool.
3. Examine the device details, including device name/description, device type, serial number, Registry Time 1 and Registry Time 2, Vendor ID and Product ID.
4. Connect the USB flash drive and locate its entry in the list, such as “Mass Storage / Generic Flash Disk USB Device”.
5. Select the device entry and open its properties to verify the serial number, assigned drive letter and last connected date and time.
6. Record the identified USB device details as forensic evidence for further analysis.
7. Explore the additional features of USBDeview, which allow users to uninstall, enable, disable or disconnect USB devices and examine devices on remote computers.

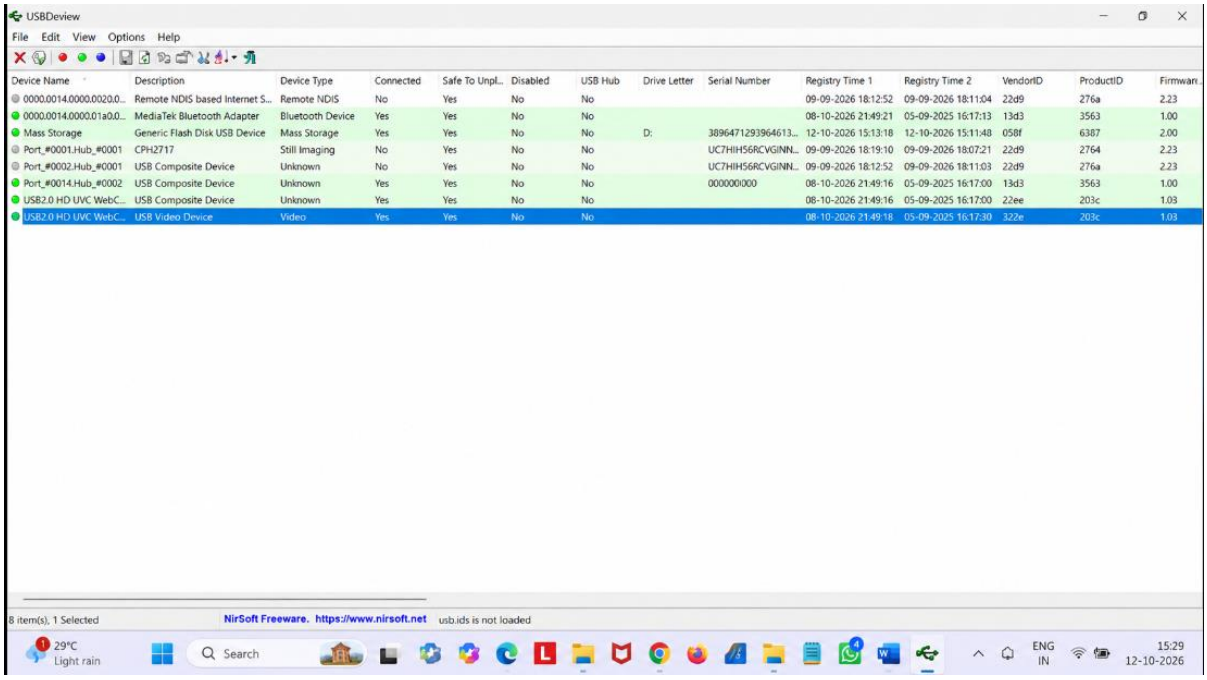



**Fig. 6.1** USBDeview – list of connected and previously used USB devices

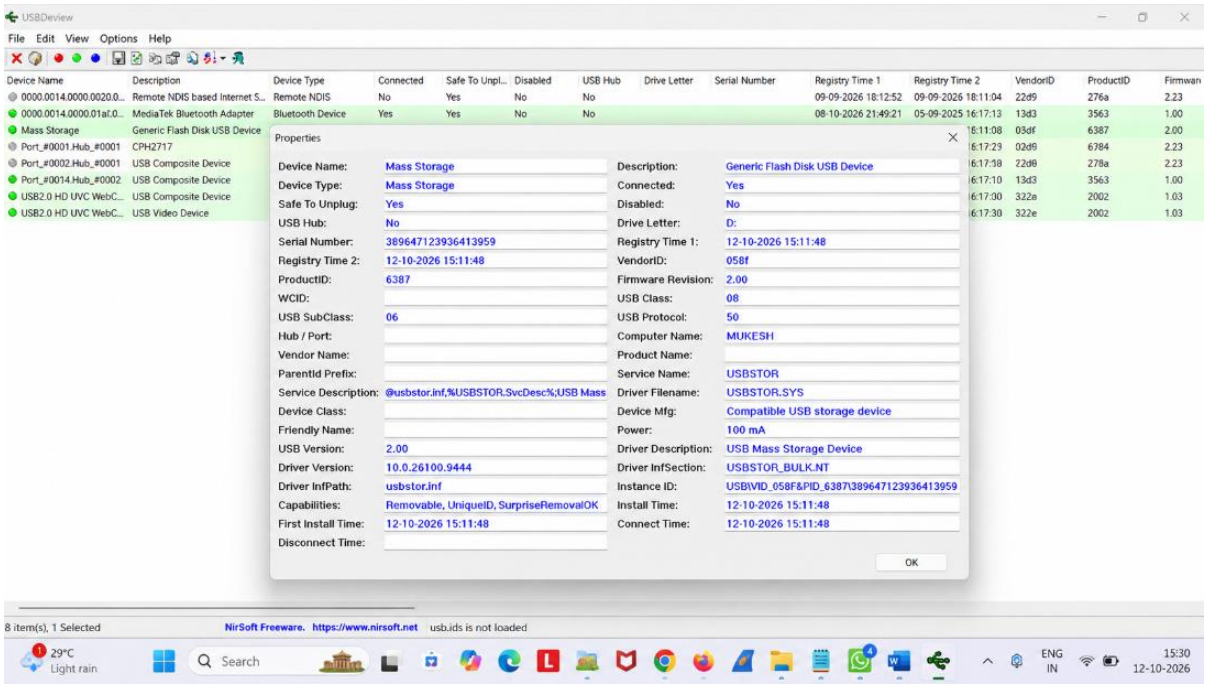



**Fig. 6.2** USBDeview – properties of the last connected USB mass-storage device

## RESULT

Thus, the last connected USB device was identified using USBDeview. The device name, serial number, VendorID/ProductID and the last connected date and time were recorded as evidence.

# EXPERIMENT – 7

## View Last Activity of Your PC

### AIM

To view the last activity performed on the system using the LastActivityView tool.

### PROCEDURE

- Download and run LastActivityView (NirSoft).
- The tool collects information from various sources on a running system and displays a log of the actions made by the user and the events that occurred on the computer.
- Observe the columns displayed – Action Time, Description, Filename, Full Path, More Information, File Extension and Data Source.
- Review the list to identify the most recent processes executed and the files or folders that were opened (for example usbdeview.exe, lastactivityview.zip, chrome.exe).
- Right-click on any record and select 'Properties' to view the detailed information about that activity, such as the exact action time and the data source (Prefetch file, registry, etc.).
- Use the collected timeline of user actions and system events for forensic analysis.

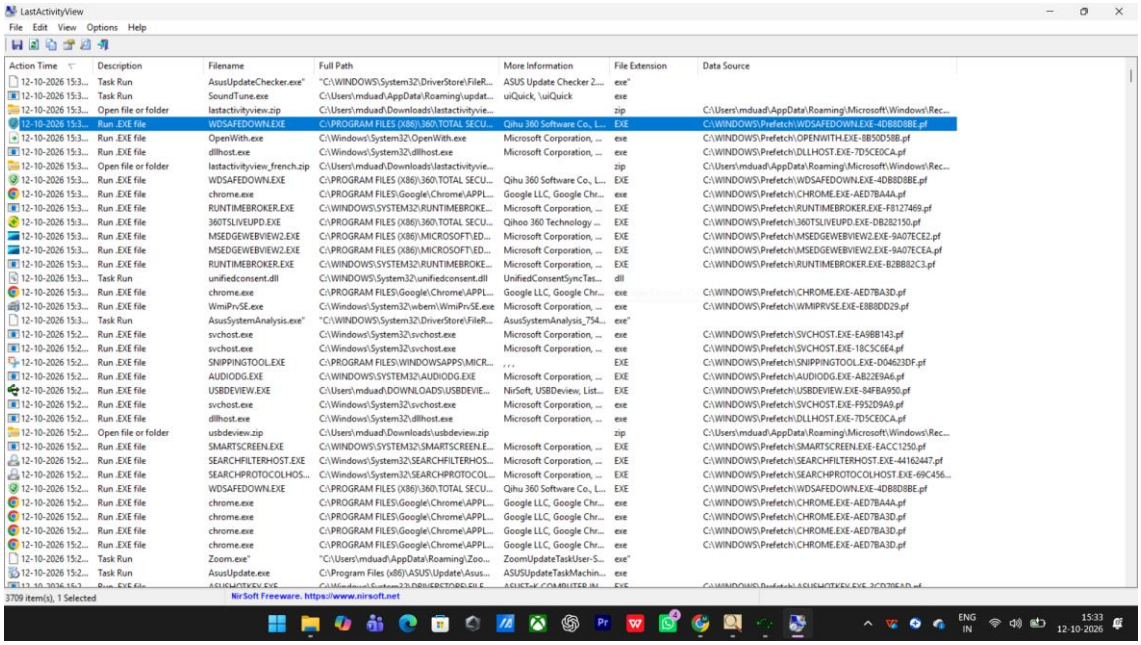



**Fig. 7.1** LastActivityView – log of recent user actions and system events

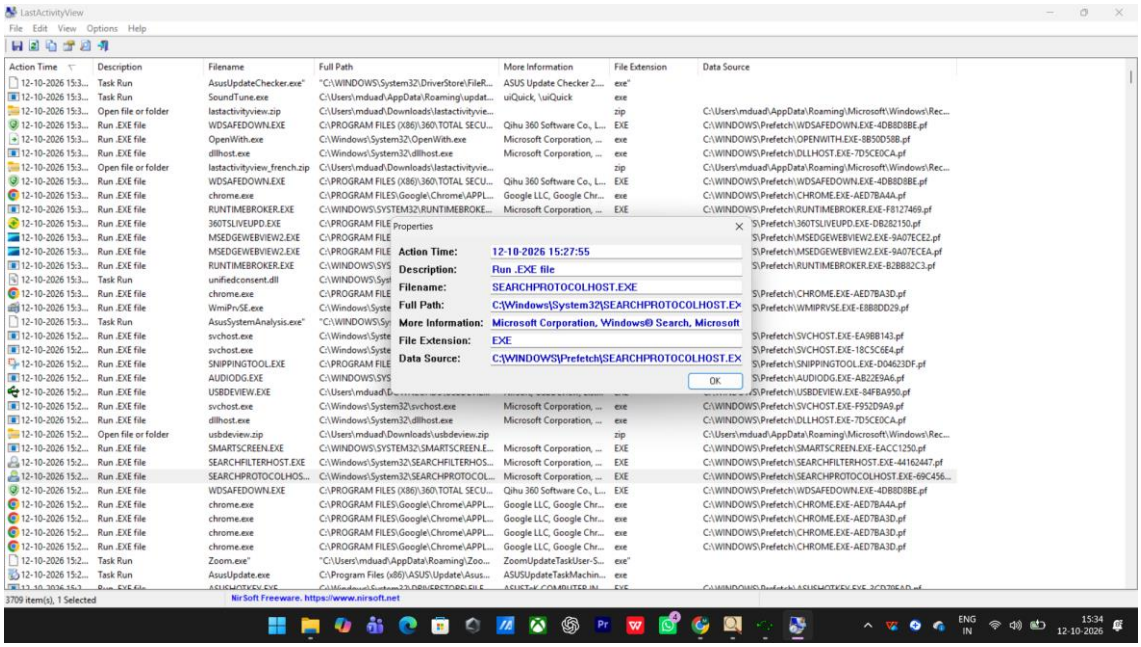



**Fig. 7.2** LastActivityView – properties of a selected activity record

## RESULT

Thus, the last activities performed on the system were viewed successfully using LastActivityView, which provided a timeline of user actions and system events for forensic analysis.

# EXPERIMENT – 8

## Extraction of Browser Artifacts

### AIM

To extract browser artifacts, such as browsing history and cache files, from the system using browser forensic tools.

### PROCEDURE

1. ChromeHistoryView: Download and run ChromeHistoryView, a utility that reads Google Chrome's history database and displays the list of previously visited web pages.

2. View browsing history: Observe details such as the URL, page title, visit date and time, number of visits, Typed Count, referrer, and Visit ID.

3. Search history: Use the Find option to search for specific keywords in the browsing history.

4. View record properties: Select a history record and examine its detailed properties, including visit count, referrer, and the Chrome user-data folder path.

5. ChromeCacheView: Run ChromeCacheView to examine cached files stored by Google Chrome. Observe details such as URL, content type, file size, and last accessed time.

6. Internet Explorer tools: Use IEHistoryView to examine browsing history and IECacheView to inspect cached files from Internet Explorer.

7. Analyze artifacts: Record the relevant browser history and cache information for forensic investigation and documentation.

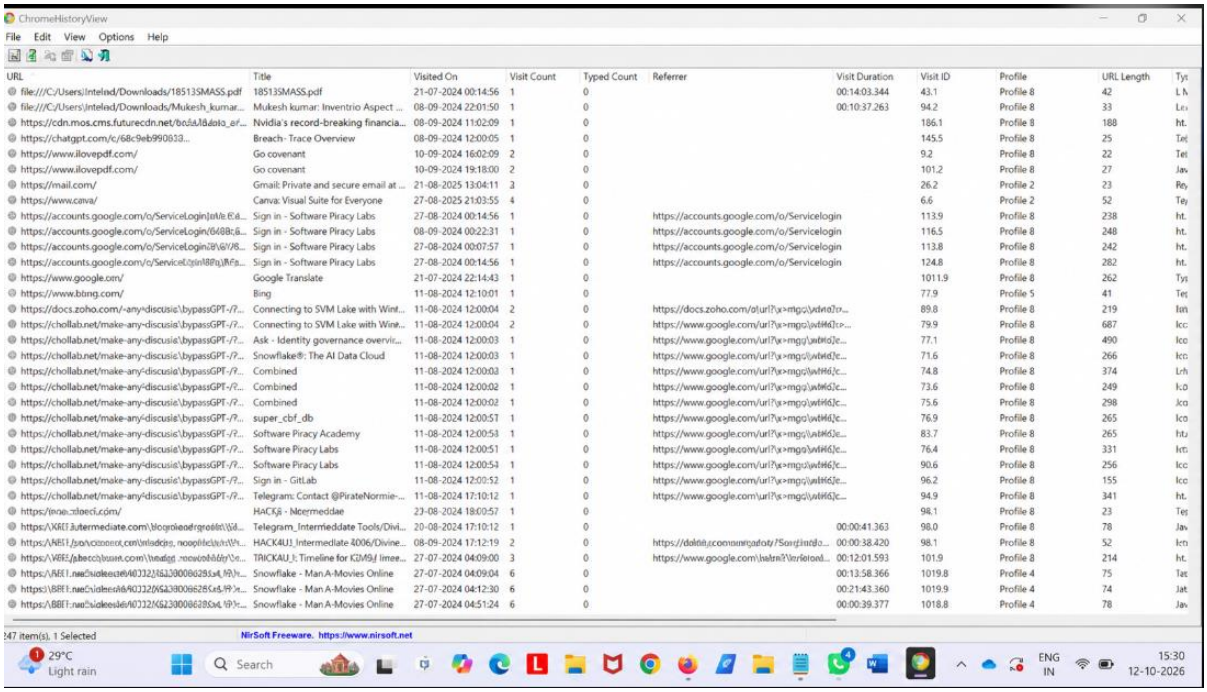



**Fig. 8.1** ChromeHistoryView – history of visited web pages

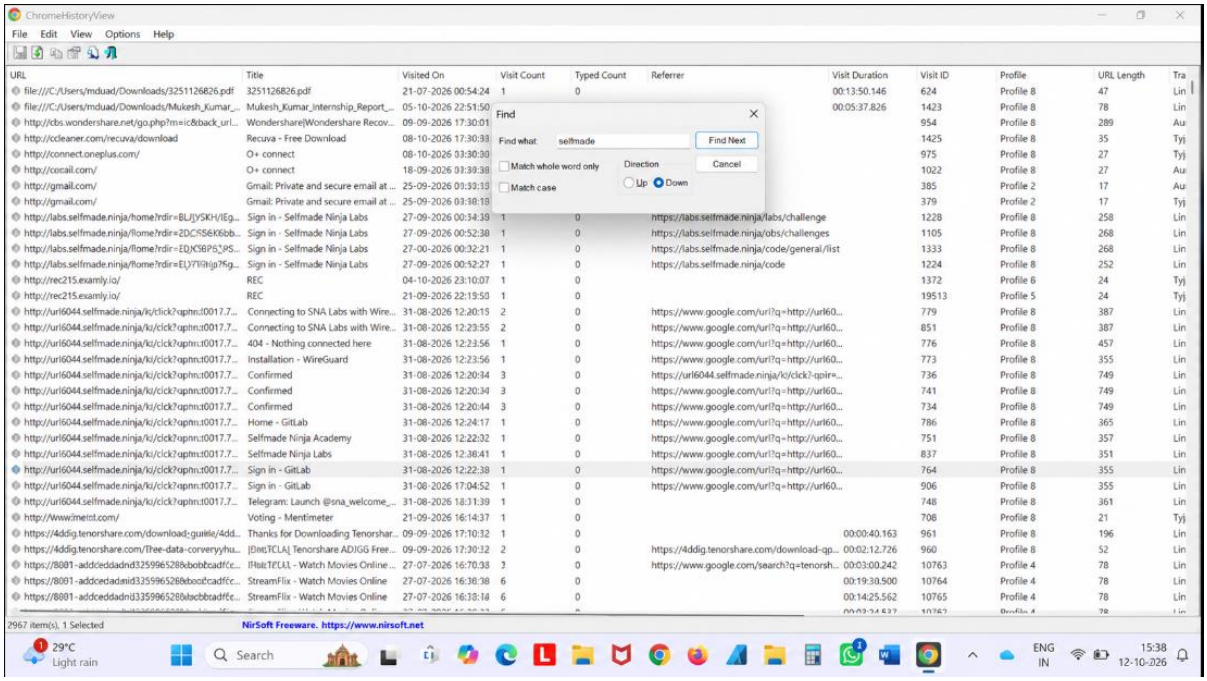



**Fig. 8.2** ChromeHistoryView – searching the history for a keyword

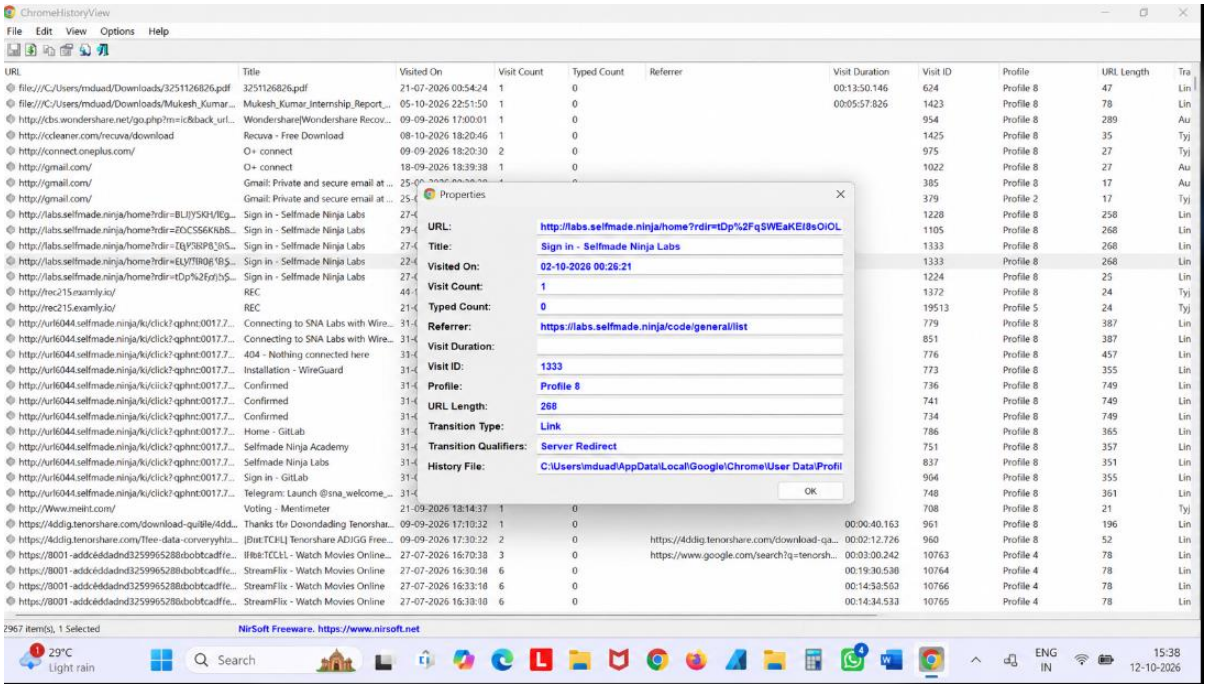



**Fig. 8.3** ChromeHistoryView – details of a selected history entry

## RESULT

Thus, the browser artifacts were extracted successfully. The list of visited web pages, along with the visit date/time, visit count and referrer, was obtained and can be used as evidence during a forensic investigation.

# EXPERIMENT – 9

## Comparison of Two Files by Compare It Software

### AIM

To compare two files for forensic investigation using the Compare It software and to generate a differences report.

### PROCEDURE

- Install the Compare It software on the system.
- Compare It displays two files side by side with coloured difference sections that simplify the analysis.
- Click on the Compare It tool; a window opens to select the files to be compared.
- Select the first file and click ‘Open’, then select the second file and click ‘Open’.
- The tool displays the changes between the two files with highlighted bars; the changes can be moved between the files with a single mouse click or keystroke.
- The files can also be edited directly inside the comparison window.
- Generate a coloured printout / report of the differences exactly as shown on the screen.

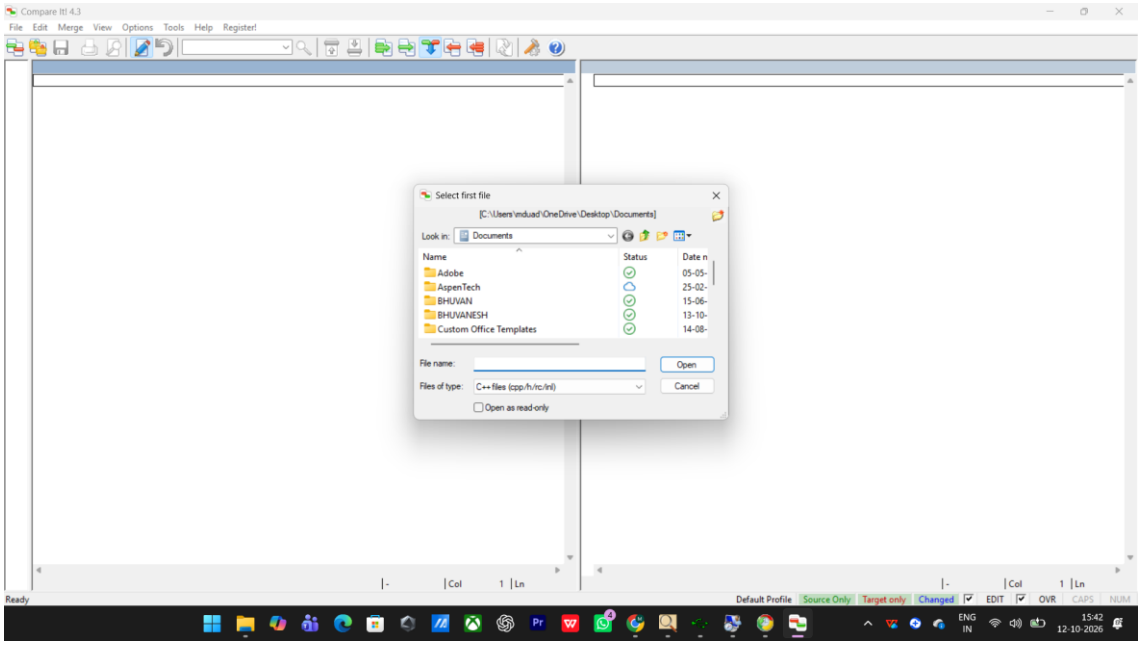



**Fig. 9.1** Compare It – selecting the first file to compare

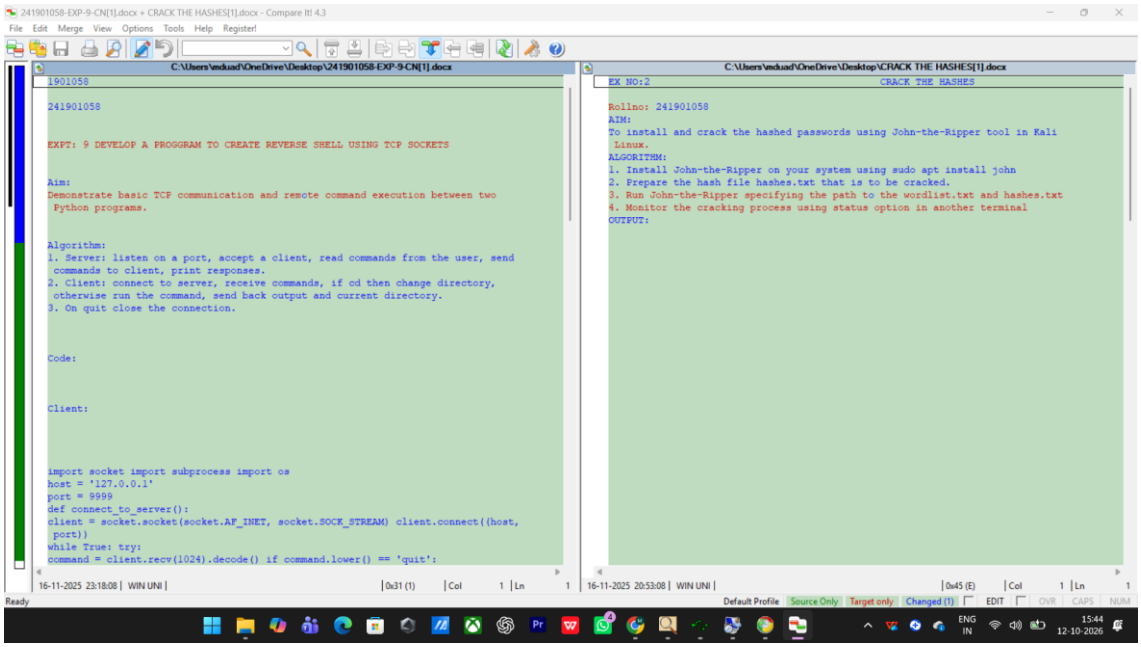



**Fig. 9.2** Compare It – side-by-side comparison showing the differences between the two files

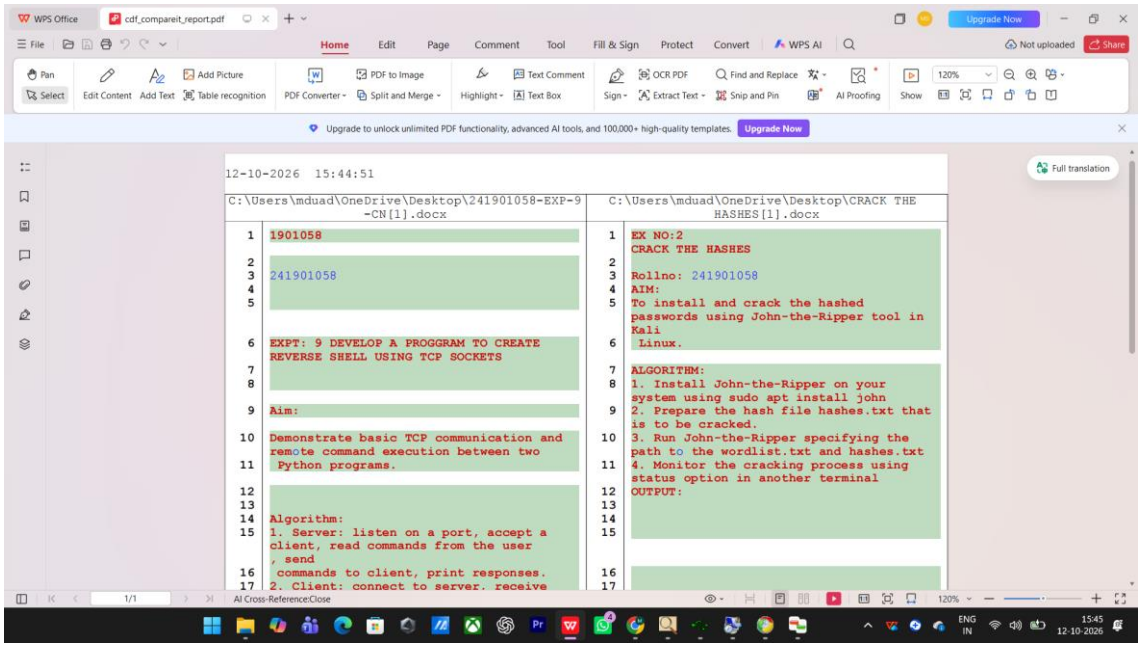



**Fig. 9.3** Generated comparison report (cat_compareit_report.pdf)

## RESULT

Thus, the two files were compared successfully using Compare It. The differences between the files were highlighted and a detailed comparison report was generated for forensic investigation.

# EXPERIMENT – 10

## Collecting Email Evidence in Victim PC

**AIM:**

To collect email evidence from a victim's PC by acquiring physical memory and extracting email artifacts.

**PROCEDURE:**

1. Memory Acquisition: Run the DumpIt tool to capture the system's RAM and generate a physical memory dump.
2. Execute dumpit.exe, type Y, and press Enter to save the memory dump as a .raw file.
3. Feature Extraction: Download and open Bulk Extractor Viewer from GitHub.
4. Select the memory dump as input, choose an output folder, and click Start to generate the report.
5. Bulk Extractor produces feature files such as email.txt, email_histogram.txt, and url.txt.
6. Open email.txt to examine extracted email addresses and email_histogram.txt to check their frequency.

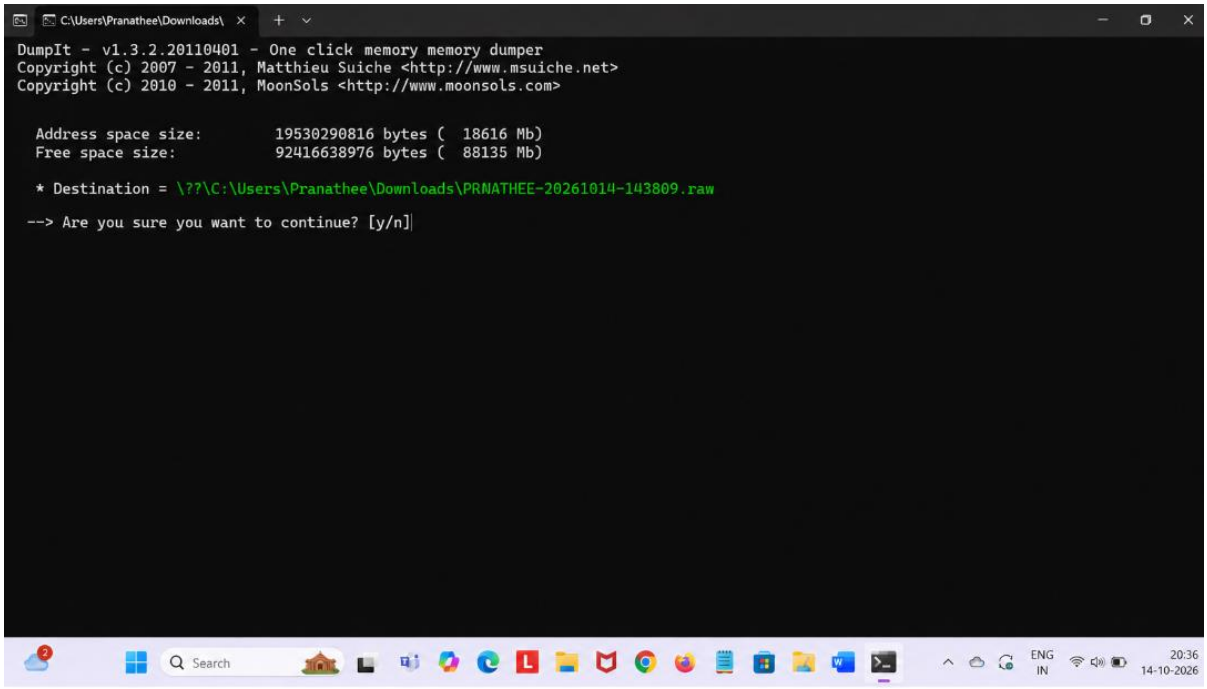



**Fig. 10.1** DumpIt – acquiring a raw physical memory dump

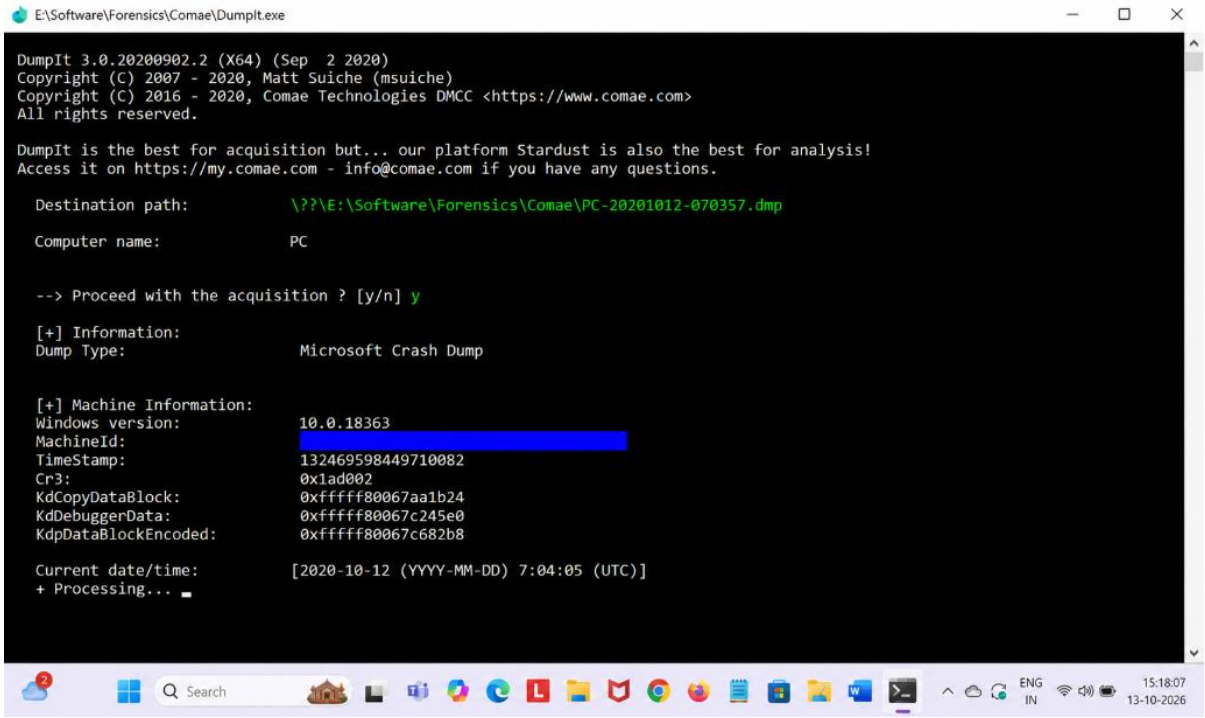



**Fig. 10.2** DumpIt – memory acquisition in progress

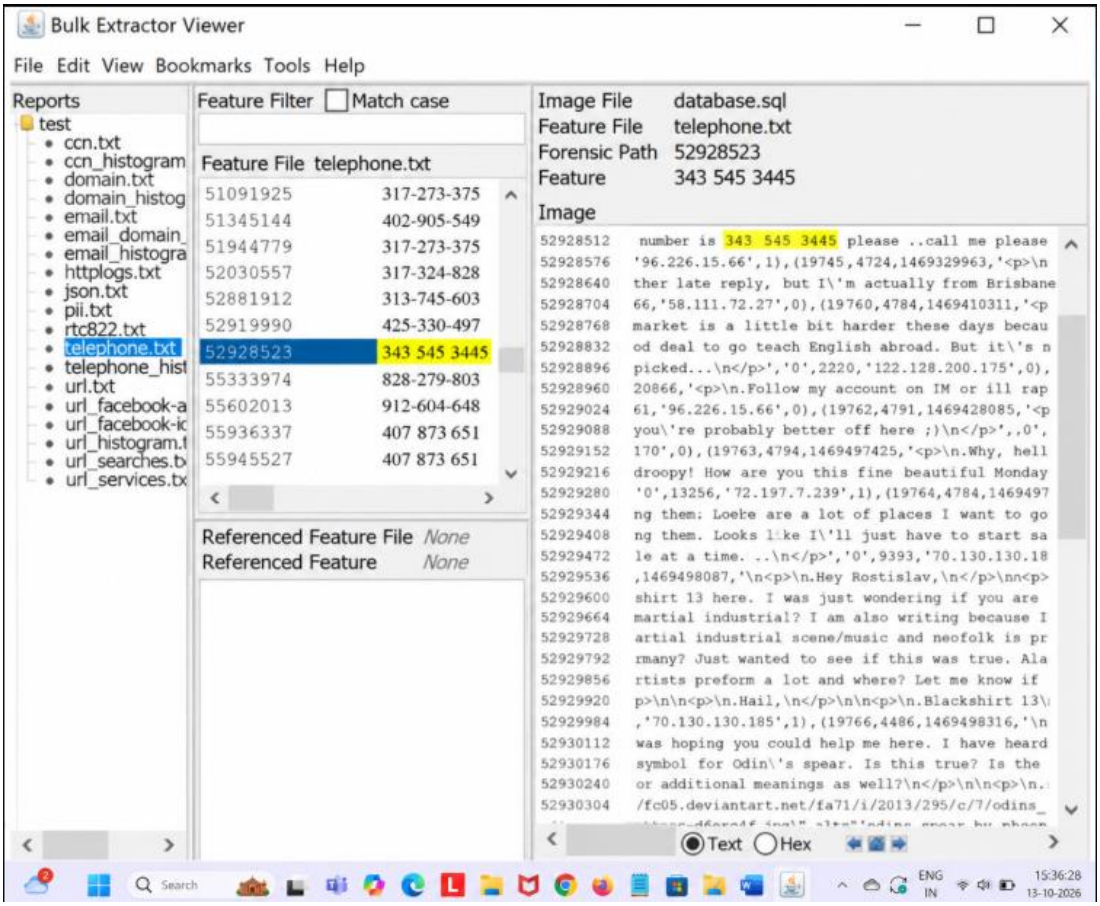



**Fig. 10.3** Bulk Extractor Viewer – extracted feature files from the memory image

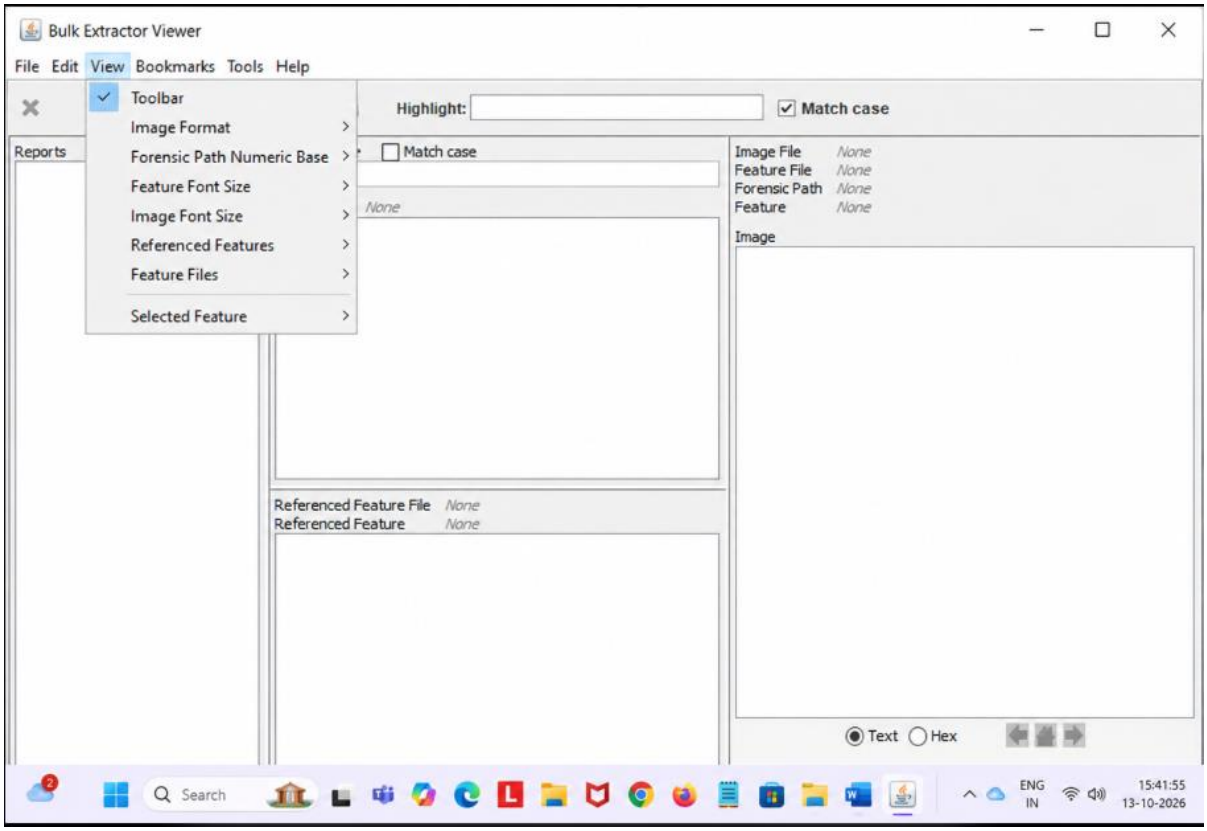



**Fig. 10.4** Bulk Extractor Viewer – viewing the extracted features

## RESULT

Thus, the e-mail evidence from the victim's PC was collected successfully. The physical memory of the victim system was acquired using DumpIt, the memory image was processed with Bulk Extractor, and the e-mail addresses and their histogram were extracted and examined.

# EXPERIMENT – 11

## Hiding and Extracting a Text File behind an Image / Audio File using Command Prompt

**AIM:**

To study the process of hiding and extracting text files behind image or audio files using the Command Prompt.

**PROCEDURE:**

1. Copy sample.txt and sample1.jpg into the same folder.
2. Open Command Prompt (Win + R → cmd) and navigate to the folder using cd.
3. Hide the text file inside the image using: copy /b sample1.jpg + sample.txt result.jpg
4. Open result.jpg with Notepad to view the hidden text at the end of the file.
5. To hide a RAR archive, use: copy /b sample1.jpg + sample.rar result.jpg
6. To hide a ZIP archive inside an audio file, use: copy /b sound.mp3 + new.zip newfile.mp3
7. In Linux, the same operation can be performed using: cat sound.mp3 secret.zip > newfile.mp3

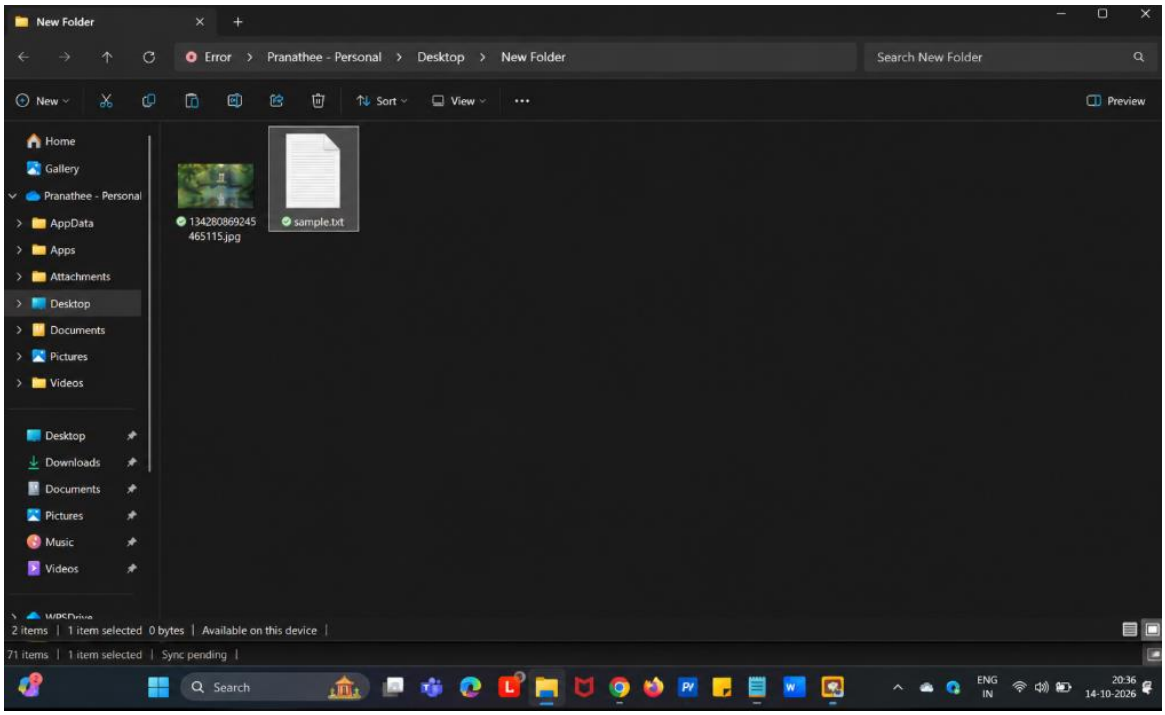



**Fig. 11.1** Placing the text file and the carrier image in a folder

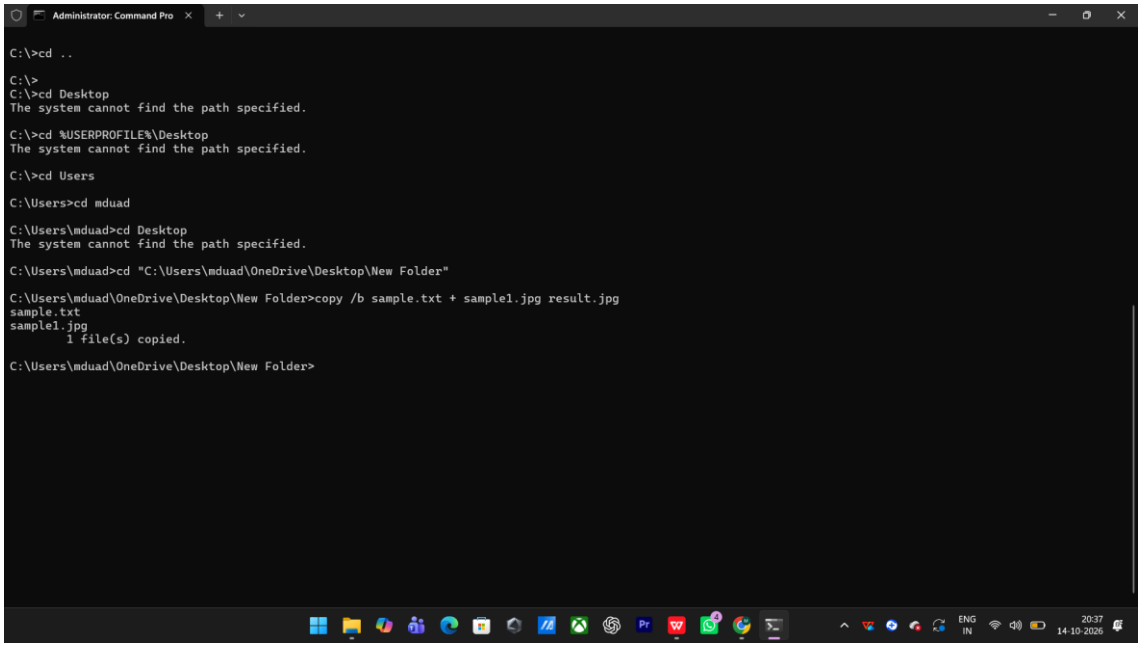



**Fig. 11.2** Command Prompt – merging the text file with the image (copy /b)

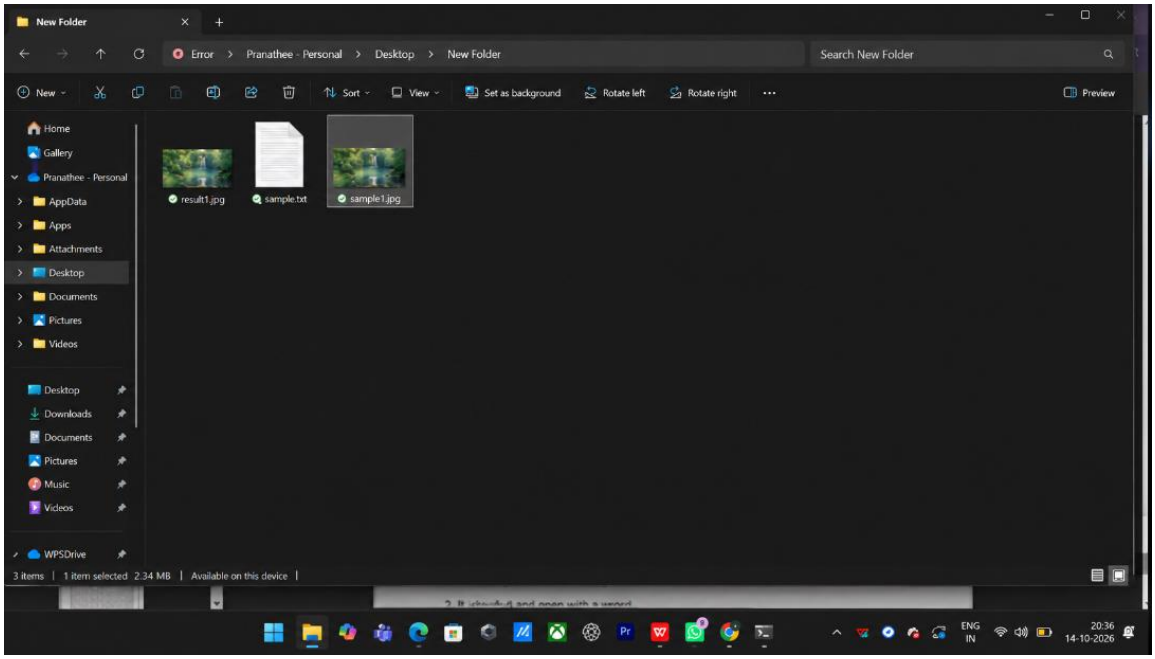



**Fig. 11.3** The output image result.jpg generated after the merge

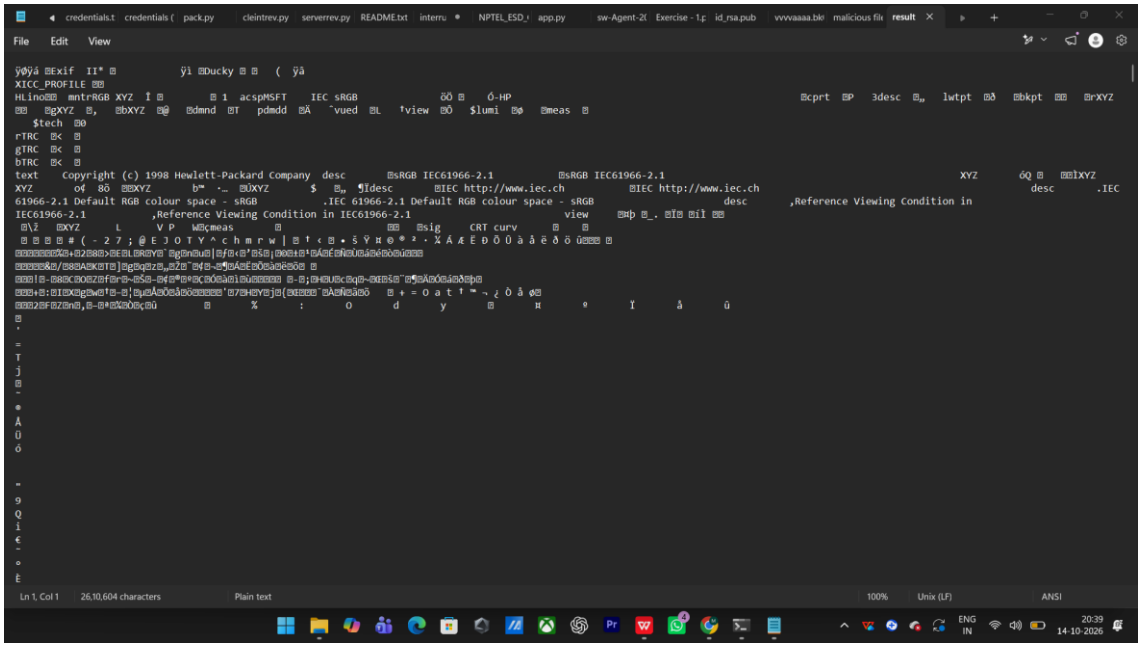



**Fig. 11.4** Opening result.jpg with Notepad to retrieve the hidden text

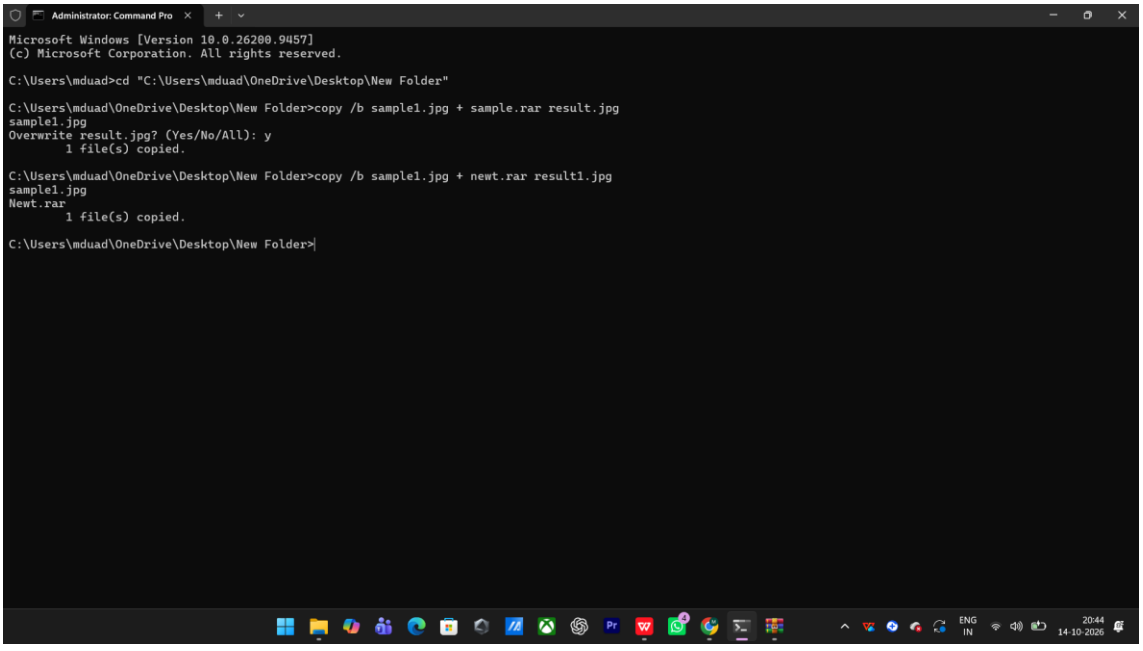



**Fig. 11.5** Command Prompt – hiding a RAR archive behind the image

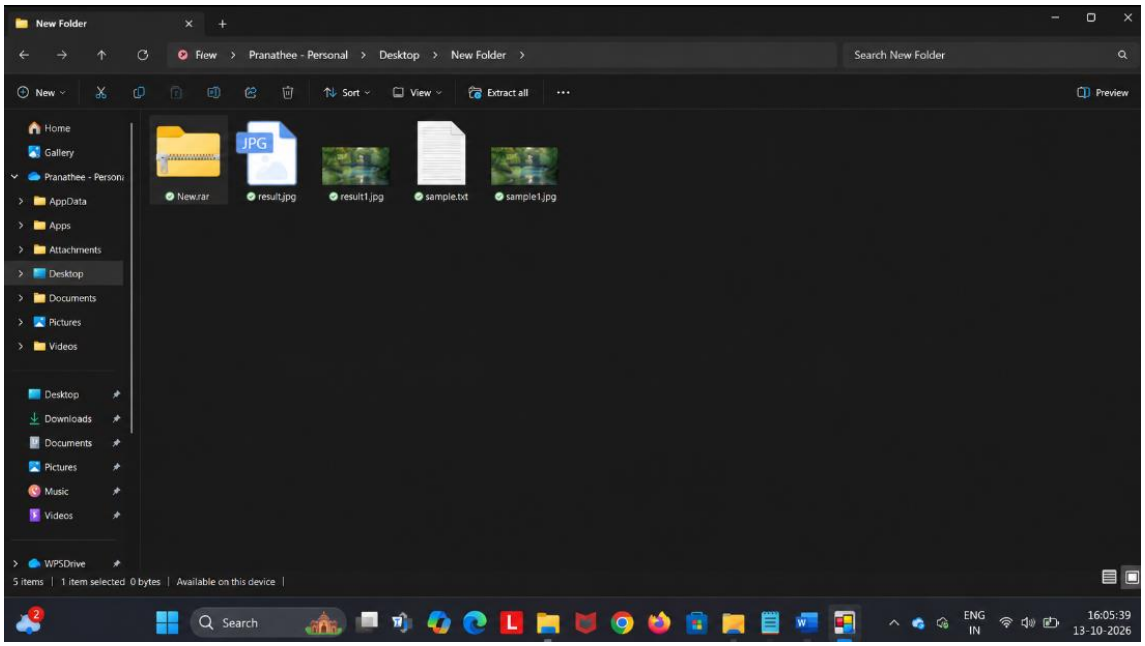



**Fig. 11.6** The output image containing the hidden RAR archive

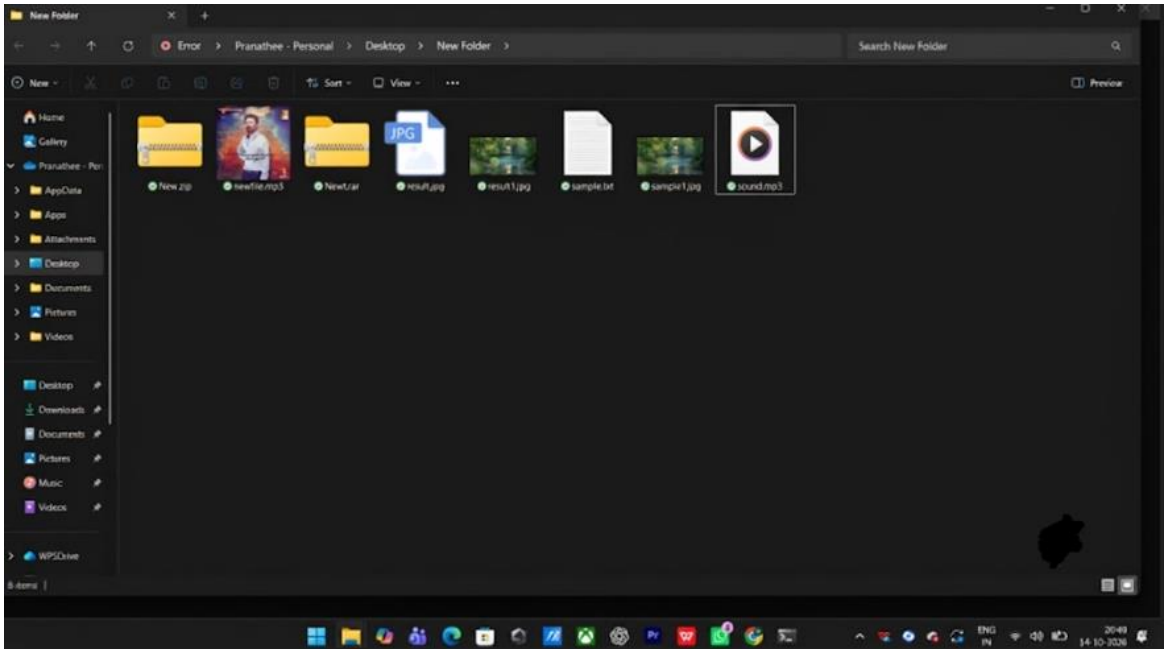



**Fig. 11.7** Files prepared for hiding data behind an audio file

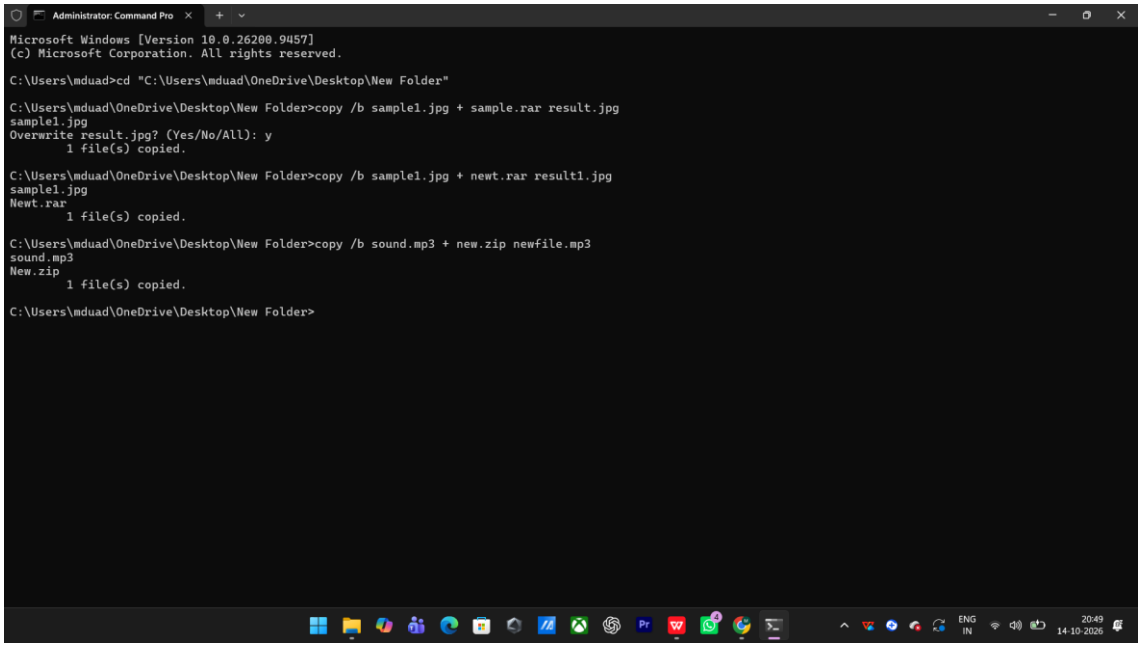



**Fig. 11.8** Command Prompt – merging the ZIP archive with the audio file

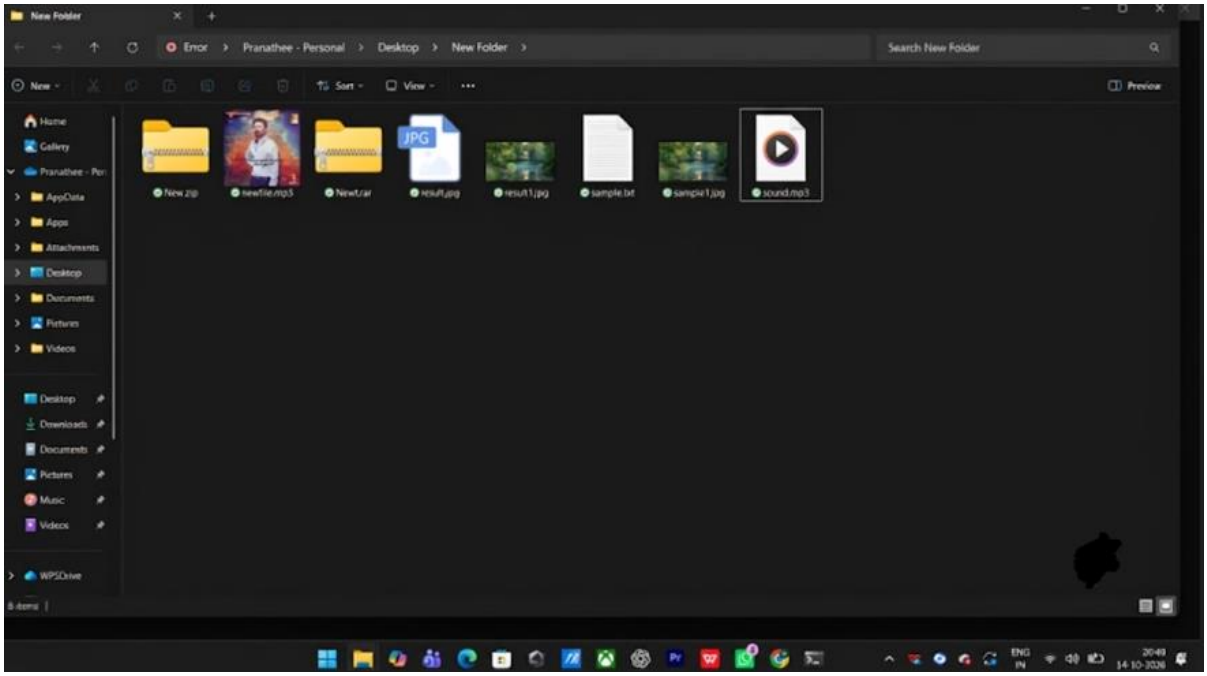



**Fig. 11.9** The output audio file newfile.mp3 containing the hidden ZIP archive

## RESULT

Thus, a text file, a RAR archive and a ZIP archive were hidden behind an image file and an audio file using the Command Prompt, and the hidden content was retrieved successfully. The output files continued to open normally as image and audio file

# EXPERIMENT – 12

## Extraction of EXIF Data from Image Files using ExifReader

### AIM

To extract the Exchangeable image file format (EXIF) data from an image file using the ExifReader software.

### INTRODUCTION

When a computer, phone or mobile device is seized for evidence, the system usually contains graphic images that may be used as evidence. Most digital devices 'stamp' information on these images that can tell us about the who, what, when and where of the picture. This information is known as EXIF data and is often useful to the forensic investigator.

EXIF is a standard set by the digital camera industry to identify formats for digital images and sound files. It includes camera settings, time, date, shutter speed, exposure, whether a flash was used, compression, the name of the camera and other information. Several applications can extract this data; for this experiment a simple Windows-based tool called ExifReader is used.

### PROCEDURE

- Download ExifReader and run the file ExifRead.exe. It opens a clean and simple GUI wizard.
- Click on the 'Open' button and browse to the picture stored on the system or media. JPEG and JPG files normally contain the maximum information.
- Once the selected picture opens, it is loaded as a thumbnail on the left and the EXIF data is displayed on the right.
- Study the extracted information such as the Make and Model of the camera, DateTime, DateTimeOriginal, Exposure Time, FNumber, Exposure Program, ISO Speed Ratings, Compressed Bits, Focal Length, Flash and the GPS location.
- Note the details that are relevant to the investigation, as EXIF data can reveal when and where an image was captured and with which device.

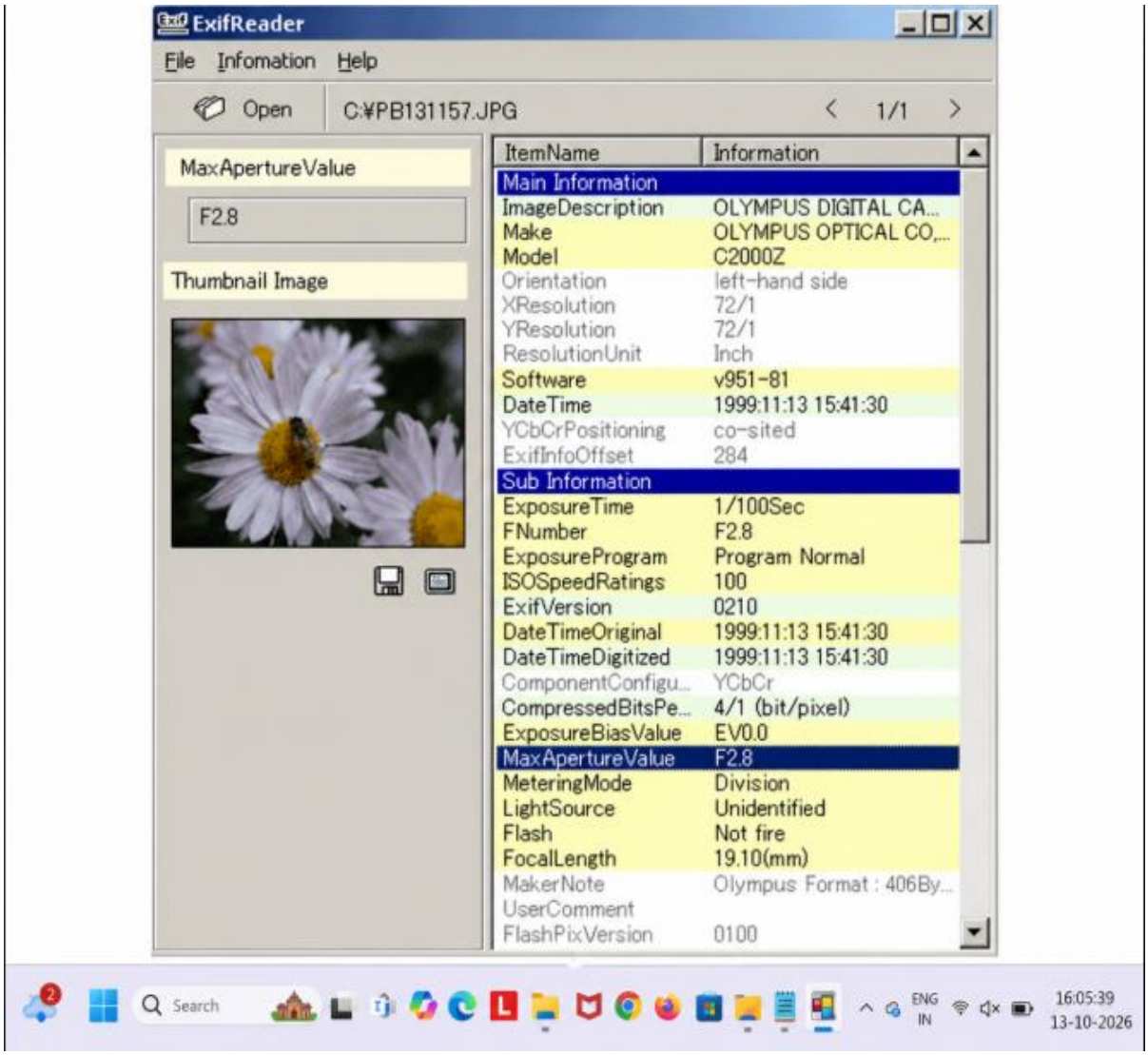



**Fig. 12.1** ExifReader – EXIF metadata extracted from a JPEG image

**RESULT**

Thus, the EXIF data was extracted successfully from the image file using the ExifReader software. The camera make and model, the date and time the picture was taken and the photographic settings were obtained, which can help the forensic investigator to link the image to its source.